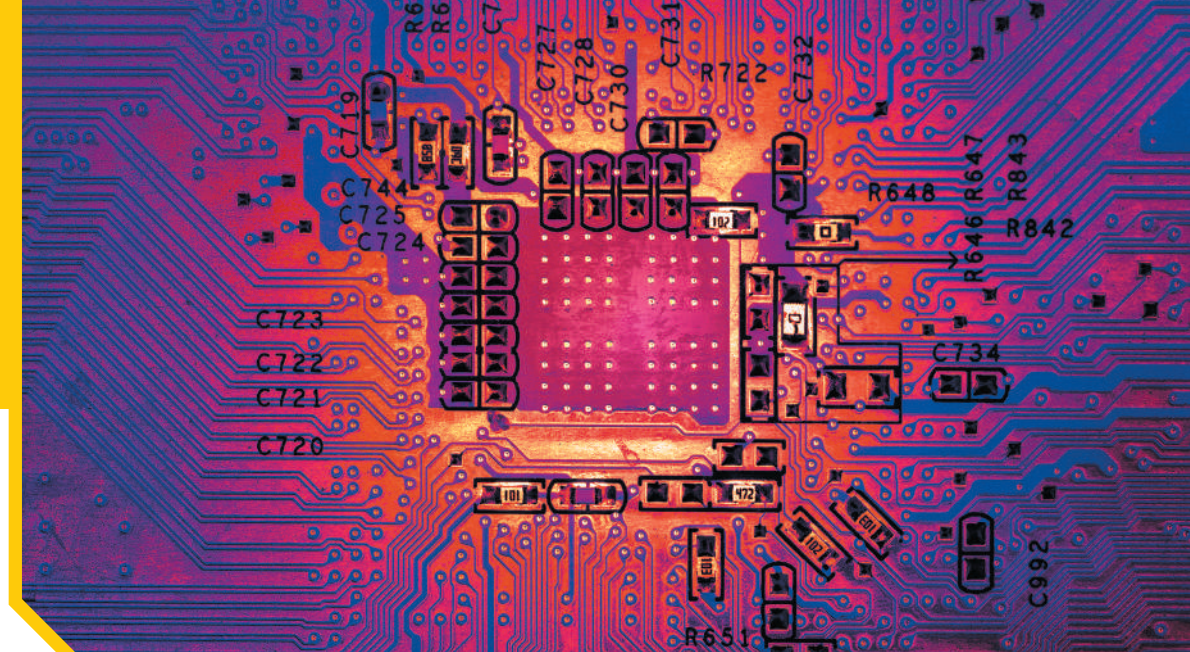


? In a complex circuit like the one on this circuit board, is it possible to connect several resistors with different resistances so that all of them have the same potential difference? (i) Yes, and the current will be the same through all of the resistors; (ii) yes, but the current may be different through different resistors; (iii) no; (iv) the answer depends on the value of the potential difference.



26 Direct-Current Circuits

If you look inside a mobile phone, a computer, or under the hood of a car, you'll find circuits of much greater complexity than the simple circuits we studied in Chapter 25. Whether connected by wires or integrated in a semiconductor chip, these circuits often include several sources, resistors, and other circuit elements interconnected in a *network*.

In this chapter we study general methods for analyzing such networks, including how to find voltages and currents of circuit elements. We'll learn how to determine the equivalent resistance for several resistors connected in series or in parallel. For more general networks we need two rules called *Kirchhoff's rules*. One is based on the principle of conservation of charge applied to a junction; the other is derived from energy conservation for a charge moving around a closed loop. We'll discuss instruments for measuring various electrical quantities. We'll also look at a circuit containing resistance and capacitance, in which the current varies with time.

Our principal concern in this chapter is with **direct-current** (dc) circuits, in which the direction of the current does not change with time. Flashlights and automobile wiring systems are examples of direct-current circuits. Household electrical power is supplied in the form of **alternating current** (ac), in which the current oscillates back and forth. The same principles for analyzing networks apply to both kinds of circuits, and we conclude this chapter with a look at household wiring systems. We'll discuss alternating-current circuits in detail in Chapter 31.

26.1 RESISTORS IN SERIES AND PARALLEL

Resistors turn up in all kinds of circuits, ranging from hair dryers and space heaters to circuits that limit or divide current or reduce or divide a voltage. Such circuits often contain several resistors, so it's appropriate to consider *combinations* of resistors. A simple example is a string of light bulbs used for holiday decorations; each bulb acts as a resistor, and from a circuit-analysis perspective the string of bulbs is simply a combination of resistors.

Suppose we have three resistors with resistances R_1 , R_2 , and R_3 . **Figure 26.1** shows four different ways in which they might be connected between points *a* and *b*. When several circuit elements such as resistors, batteries, and motors are connected in sequence as

LEARNING OUTCOMES

In this chapter, you'll learn...

- 26.1** How to analyze circuits with multiple resistors in series or parallel.
- 26.2** Rules that you can apply to any circuit with more than one loop.
- 26.3** How to use an ammeter, voltmeter, ohmmeter, or potentiometer in a circuit.
- 26.4** How to analyze circuits that include both a resistor and a capacitor.
- 26.5** How electric power is distributed in the home.

You'll need to review...

- 24.2** Capacitors in series and parallel.
- 25.4** Current, ammeters, and voltmeters.
- 25.5** Power in a circuit.

in Fig. 26.1a, with only a single current path between the points, we say that they are connected in **series**. We studied *capacitors* in series in Section 24.2; we found that, because of conservation of charge, capacitors in series all have the same charge if they are initially uncharged. In circuits we're often more interested in the *current*, which is charge flow per unit time.

The resistors in Fig. 26.1b are said to be connected in **parallel** between points *a* and *b*. Each resistor provides an alternative path between the points. For circuit elements that are connected in parallel, the *potential difference* is the same across each element. We studied capacitors in parallel in Section 24.2.

In Fig. 26.1c, resistors R_2 and R_3 are in parallel, and this combination is in series with R_1 . In Fig. 26.1d, R_2 and R_3 are in series, and this combination is in parallel with R_1 .

For any combination of resistors we can always find a *single* resistor that could replace the combination and result in the same total current and potential difference. For example, a string of holiday light bulbs could be replaced by a single, appropriately chosen light bulb that would draw the same current and have the same potential difference between its terminals as the original string of bulbs. The resistance of this single resistor is called the **equivalent resistance** of the combination. If any one of the networks in Fig. 26.1 were replaced by its equivalent resistance R_{eq} , we could write

$$V_{ab} = IR_{\text{eq}} \quad \text{or} \quad R_{\text{eq}} = \frac{V_{ab}}{I}$$

where V_{ab} is the potential difference between terminals *a* and *b* of the network and I is the current at point *a* or *b*. To compute an equivalent resistance, we assume a potential difference V_{ab} across the actual network, compute the corresponding current I , and take the ratio V_{ab}/I .

Resistors in Series

We can derive general equations for the equivalent resistance of a series or parallel combination of resistors. In Fig. 26.1a, the three resistors are in *series*, so the current I is the same in all of them. (Recall from Section 25.4 that current is *not* “used up” as it passes through a circuit.) Applying $V = IR$ to each resistor, we have

$$V_{ax} = IR_1 \quad V_{xy} = IR_2 \quad V_{yb} = IR_3$$

The potential differences across each resistor need not be the same (except for the special case in which all three resistances are equal). The potential difference V_{ab} across the entire combination is the sum of these individual potential differences:

$$V_{ab} = V_{ax} + V_{xy} + V_{yb} = I(R_1 + R_2 + R_3)$$

and so

$$\frac{V_{ab}}{I} = R_1 + R_2 + R_3$$

The ratio V_{ab}/I is, by definition, the equivalent resistance R_{eq} . Therefore

$$R_{\text{eq}} = R_1 + R_2 + R_3$$

It is easy to generalize this to *any* number of resistors:

Resistors
in series:

$R_{\text{eq}} = R_1 + R_2 + R_3 + \cdots$

(26.1)

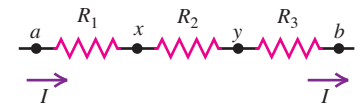
Equivalent resistance
of series combination

Resistances of
individual resistors

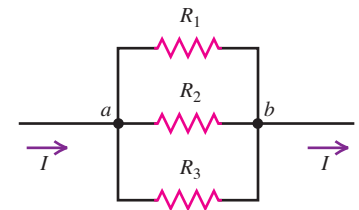
The equivalent resistance of a series combination equals the sum of the individual resistances. The equivalent resistance is *greater than* any individual resistance.

Figure 26.1 Four different ways of connecting three resistors.

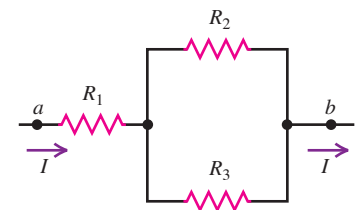
(a) R_1 , R_2 , and R_3 in series



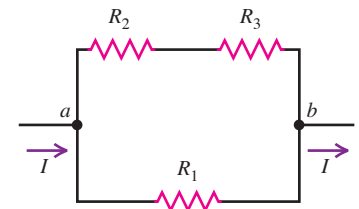
(b) R_1 , R_2 , and R_3 in parallel



(c) R_1 in series with parallel combination of R_2 and R_3



(d) R_1 in parallel with series combination of R_2 and R_3



CAUTION Resistors vs. capacitors in series Don't confuse *resistors* in series with *capacitors* in series. Resistors in series add *directly* [Eq. (26.1)] because the voltage across each is directly proportional to its resistance and to the common current. Capacitors in series [Eq. (24.5)] add *reciprocally*; the voltage across each is directly proportional to the common charge but *inversely* proportional to the individual capacitance. ■

Figure 26.2 A car's headlights and taillights are connected in parallel. Hence each light is exposed to the full potential difference supplied by the car's electrical system, giving maximum brightness. Another advantage is that if one headlight or taillight burns out, the other one keeps shining (see Example 26.2).



Resistors in Parallel

If the resistors are in *parallel*, as in Fig. 26.1b, the current through each resistor need **?** not be the same. But the potential difference between the terminals of each resistor must be the same and equal to V_{ab} (**Fig. 26.2**). (Remember that the potential difference between any two points does not depend on the path taken between the points.) Let's call the currents in the three resistors I_1 , I_2 , and I_3 . Then from $I = V/R$,

$$I_1 = \frac{V_{ab}}{R_1} \quad I_2 = \frac{V_{ab}}{R_2} \quad I_3 = \frac{V_{ab}}{R_3}$$

In general, the current is different through each resistor. Because charge neither accumulates at nor drains out of point a , the total current I must equal the sum of the three currents in the resistors:

$$I = I_1 + I_2 + I_3 = V_{ab} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$$

or

$$\frac{I}{V_{ab}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

But by the definition of the equivalent resistance R_{eq} , $I/V_{ab} = 1/R_{eq}$, so

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Again it is easy to generalize to *any* number of resistors in parallel:

$$\text{Resistors in parallel:} \quad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots \quad (26.2)$$

Equivalent resistance of parallel combination Resistances of individual resistors

The reciprocal of the equivalent resistance of a parallel combination equals the sum of the reciprocals of the individual resistances. The equivalent resistance is always *less than* any individual resistance.

CAUTION Resistors vs. capacitors in parallel Note the differences between *resistors* in parallel and *capacitors* in parallel. Resistors in parallel add *reciprocally* [Eq. (26.2)] because the current in each is proportional to the common voltage across them and *inversely* proportional to the resistance of each. Capacitors in parallel add *directly* [Eq. (24.7)] because the charge on each is proportional to the common voltage across them and *directly* proportional to the capacitance of each. ■

For the special case of *two* resistors in parallel,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_1 + R_2}{R_1 R_2} \quad \text{and}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} \quad (\text{two resistors in parallel}) \quad (26.3)$$

Because $V_{ab} = I_1 R_1 = I_2 R_2$, it follows that

$$\frac{I_1}{I_2} = \frac{R_2}{R_1} \quad (\text{two resistors in parallel}) \quad (26.4)$$

Thus the currents carried by two resistors in parallel are *inversely proportional* to their resistances. More current goes through the path of least resistance.

PROBLEM-SOLVING STRATEGY 26.1 Resistors in Series and Parallel

IDENTIFY the relevant concepts: As in Fig. 26.1, many resistor networks are made up of resistors in series, in parallel, or a combination thereof. Such networks can be replaced by a single equivalent resistor. The logic is similar to that of Problem-Solving Strategy 24.1 for networks of capacitors.

SET UP the problem using the following steps:

1. Make a drawing of the resistor network.
2. Identify groups of resistors connected in series or parallel.
3. Identify the target variables. They could include the equivalent resistance of the network, the potential difference across each resistor, or the current through each resistor.

EXECUTE the solution as follows:

1. Use Eq. (26.1) or (26.2), respectively, to find the equivalent resistance for series or parallel combinations.
2. If the network is more complex, try reducing it to series and parallel combinations. For example, in Fig. 26.1c we first replace the

parallel combination of R_2 and R_3 with its equivalent resistance; this then forms a series combination with R_1 . In Fig. 26.1d, the combination of R_2 and R_3 in series forms a parallel combination with R_1 .

3. Keep in mind that the total potential difference across resistors connected in series is the sum of the individual potential differences. The potential difference across resistors connected in parallel is the same for every resistor and equals the potential difference across the combination.
4. The current through resistors connected in series is the same through every resistor and equals the current through the combination. The total current through resistors connected in parallel is the sum of the currents through the individual resistors.

EVALUATE your answer: Check whether your results are consistent. The equivalent resistance of resistors connected in series should be greater than that of any individual resistor; that of resistors in parallel should be less than that of any individual resistor.

EXAMPLE 26.1 Equivalent resistance

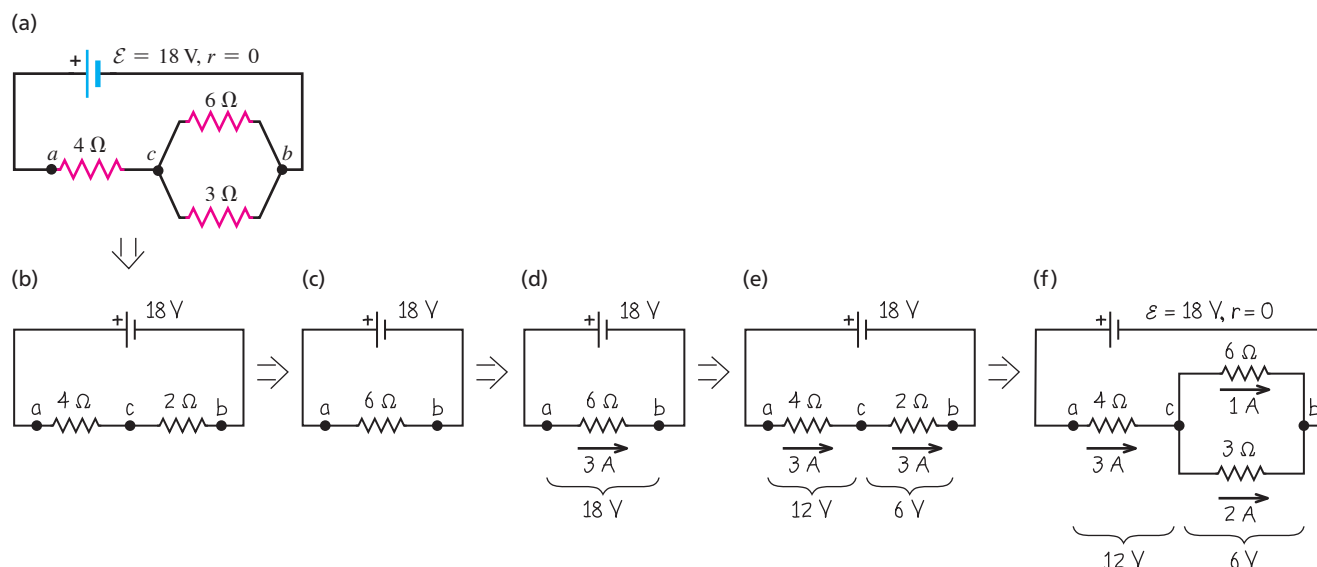
WITH VARIATION PROBLEMS

Find the equivalent resistance of the network in **Fig. 26.3a** and the current in each resistor. The source of emf has negligible internal resistance.

IDENTIFY and SET UP This network of three resistors is a *combination* of series and parallel resistances, as in Fig. 26.1c. We determine the equivalent resistance of the parallel $6\ \Omega$ and $3\ \Omega$ resistors, and then

that of their series combination with the $4\ \Omega$ resistor: This is the equivalent resistance R_{eq} of the network as a whole. We then find the current in the emf, which is the same as that in the $4\ \Omega$ resistor. The potential difference is the same across each of the parallel $6\ \Omega$ and $3\ \Omega$ resistors; we use this to determine how the current is divided between these.

Figure 26.3 Steps in reducing a combination of resistors to a single equivalent resistor and finding the current in each resistor.



Continued

EXECUTE Figures 26.3b and 26.3c show successive steps in reducing the network to a single equivalent resistance R_{eq} . From Eq. (26.2), the $6\ \Omega$ and $3\ \Omega$ resistors in parallel in Fig. 26.3a are equivalent to the single $2\ \Omega$ resistor in Fig. 26.3b:

$$\frac{1}{R_{6+3\ \Omega}} = \frac{1}{6\ \Omega} + \frac{1}{3\ \Omega} = \frac{1}{2\ \Omega}$$

[Equation (26.3) gives the same result.] From Eq. (26.1) the series combination of this $2\ \Omega$ resistor with the $4\ \Omega$ resistor is equivalent to the single $6\ \Omega$ resistor in Fig. 26.3c.

We reverse these steps to find the current in each resistor of the original network. In the circuit shown in Fig. 26.3d (identical to Fig. 26.3c), the current is $I = V_{ab}/R = (18\ \text{V})/(6\ \Omega) = 3\ \text{A}$. So the current in the $4\ \Omega$ and $2\ \Omega$ resistors in Fig. 26.3e (identical to Fig. 26.3b) is also $3\ \text{A}$. The potential difference V_{cb} across the $2\ \Omega$ resistor is therefore $V_{cb} = IR = (3\ \text{A})(2\ \Omega) = 6\ \text{V}$. This potential difference must also be $6\ \text{V}$ in Fig. 26.3f (identical to Fig. 26.3a). From $I = V_{cb}/R$, the

currents in the $6\ \Omega$ and $3\ \Omega$ resistors in Fig. 26.3f are, respectively, $(6\ \text{V})/(6\ \Omega) = 1\ \text{A}$ and $(6\ \text{V})/(3\ \Omega) = 2\ \text{A}$.

EVALUATE Note that for the two resistors in parallel between points c and b in Fig. 26.3f, there is twice as much current through the $3\ \Omega$ resistor as through the $6\ \Omega$ resistor; more current goes through the path of least resistance, in accordance with Eq. (26.4). Note also that the total current through these two resistors is $3\ \text{A}$, the same as it is through the $4\ \Omega$ resistor between points a and c .

KEYCONCEPT When resistors are combined in *series*, the current is the same through each resistor but the potential differences across the resistors may not be the same. The equivalent resistance of a series combination equals the sum of the individual resistances. When resistors are combined in *parallel*, the potential difference across each resistor is the same but the currents through the resistors may not be the same. The reciprocal of the equivalent resistance of a parallel combination equals the sum of the reciprocals of the individual resistances.

EXAMPLE 26.2 Series versus parallel combinations

WITH VARIATION PROBLEMS

Two identical incandescent light bulbs, each with resistance $R = 2\ \Omega$, are connected to a source with $\mathcal{E} = 8\ \text{V}$ and negligible internal resistance. Find the current through each bulb, the potential difference across each bulb, and the power delivered to each bulb and to the entire network if the bulbs are connected (a) in series and (b) in parallel. (c) Suppose one of the bulbs burns out; that is, its filament breaks and current can no longer flow through it. What happens to the other bulb in the series case? In the parallel case?

IDENTIFY and SET UP The light bulbs are just resistors in simple series and parallel connections (Figs. 26.4a and 26.4b). Once we find the current I through each bulb, we can find the power delivered to each bulb by using Eq. (25.18), $P = I^2R = V^2/R$.

EXECUTE (a) From Eq. (26.1) the equivalent resistance of the two bulbs between points a and c in Fig. 26.4a is $R_{\text{eq}} = 2R = 2(2\ \Omega) = 4\ \Omega$. In series, the current is the same through each bulb:

$$I = \frac{V_{ac}}{R_{\text{eq}}} = \frac{8\ \text{V}}{4\ \Omega} = 2\ \text{A}$$

Since the bulbs have the same resistance, the potential difference is the same across each bulb:

$$V_{ab} = V_{bc} = IR = (2\ \text{A})(2\ \Omega) = 4\ \text{V}$$

From Eq. (25.18), the power delivered to each bulb is

$$P = I^2R = (2\ \text{A})^2(2\ \Omega) = 8\ \text{W} \quad \text{or}$$

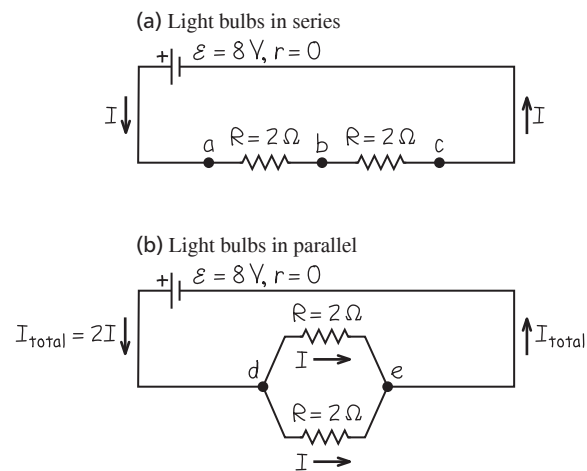
$$P = \frac{V_{ab}^2}{R} = \frac{V_{bc}^2}{R} = \frac{(4\ \text{V})^2}{2\ \Omega} = 8\ \text{W}$$

The total power delivered to both bulbs is $P_{\text{tot}} = 2P = 16\ \text{W}$.

(b) If the bulbs are in parallel, as in Fig. 26.4b, the potential difference V_{de} across each bulb is the same and equal to $8\ \text{V}$, the terminal voltage of the source. Hence the current through each light bulb is

$$I = \frac{V_{de}}{R} = \frac{8\ \text{V}}{2\ \Omega} = 4\ \text{A}$$

Figure 26.4 Our sketches for this problem.



and the power delivered to each bulb is

$$P = I^2R = (4\ \text{A})^2(2\ \Omega) = 32\ \text{W} \quad \text{or}$$

$$P = \frac{V_{de}^2}{R} = \frac{(8\ \text{V})^2}{2\ \Omega} = 32\ \text{W}$$

Both the potential difference across each bulb and the current through each bulb are twice as great as in the series case. Hence the power delivered to each bulb is *four* times greater, and each bulb is brighter.

The total power delivered to the parallel network is $P_{\text{total}} = 2P = 64\ \text{W}$, four times greater than in the series case. The increased power compared to the series case isn't obtained "for free"; energy is extracted from the source four times more rapidly in the parallel case than in the series case. If the source is a battery, it will be used up four times as fast.

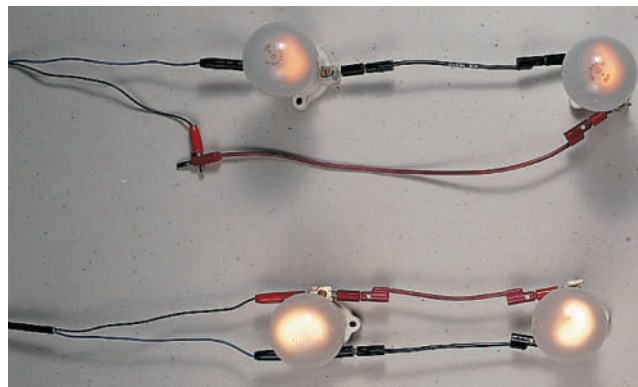
(c) In the series case the same current flows through both bulbs. If one bulb burns out, there will be no current in the circuit, and neither bulb will glow.

In the parallel case the potential difference across either bulb is unchanged if a bulb burns out. The current through the functional bulb and the power delivered to it are unchanged.

EVALUATE Our calculation isn't completely accurate, because the resistance $R = V/I$ of real light bulbs depends on the potential difference V across the bulb. That's because the filament resistance increases with increasing operating temperature and therefore with increasing V . But bulbs connected in series across a source do in fact glow less brightly than when connected in parallel across the same source (Fig. 26.5).

KEYCONCEPT Two or more resistors connected in parallel to a source each have more current through them and each draw more power than if they were connected in series to the same source.

Figure 26.5 When connected to the same source, two incandescent light bulbs in series (shown at top) draw less power and glow less brightly than when they are in parallel (shown at bottom).



TEST YOUR UNDERSTANDING OF SECTION 26.1 Suppose all three of the resistors shown in Fig. 26.1 have the same resistance, so $R_1 = R_2 = R_3 = R$. Rank the four arrangements shown in parts (a)–(d) of Fig. 26.1 in order of their equivalent resistance, from highest to lowest.

ANSWER

(a), (c), (d), (b) Here's why: The three resistors in Fig. 26.1a are in series, so $R_{eq} = R + R + R = 3R$. In Fig. 26.1b the three resistors are in parallel, so $1/R_{eq} = 1/R + 1/R + 1/R = 3/R$ and $R_{eq} = R/3$. In Fig. 26.1c the second and third resistors are in parallel, so their equivalent resistance R_{23} is given by $1/R_{23} = 1/R + 1/R = 2/R$; hence $R_{23} = R/2$. This combination is in series with the first resistor, so the three resistors together have equivalent resistance $R_{eq} = R + R/2 = 3R/2$. In Fig. 26.1d the second and third resistors are in series, so their equivalent resistance is $R_{23} = R + R = 2R$. This combination is in parallel with the first resistor, so the equivalent resistance of the three-resistor combination is given by $1/R_{eq} = 1/R + 1/2R = 3/2R$. Hence $R_{eq} = 2R/3$.

26.2 KIRCHHOFF'S RULES

Many practical resistor networks cannot be reduced to simple series-parallel combinations. Figure 26.6a shows a dc power supply with emf $\mathcal{E}_1$ charging a battery with a smaller emf $\mathcal{E}_2$ and feeding current to a light bulb with resistance R . Figure 26.6b is a “bridge” circuit, used in many different types of measurement and control systems. (Problem 26.74 describes one important application of a “bridge” circuit.) To analyze these networks, we'll use the techniques developed by the German physicist Gustav Robert Kirchhoff (1824–1887).

First, here are two terms that we'll use often. A **junction** in a circuit is a point where three or more conductors meet. A **loop** is any closed conducting path. In Fig. 26.6a points a and b are junctions, but points c and d are not; in Fig. 26.6b points a , b , c , and d are junctions, but points e and f are not. The blue lines in Figs. 26.6a and 26.6b show some possible loops in these circuits.

Kirchhoff's rules are the following two statements:

Kirchhoff's junction rule
(valid at any junction):

The sum of the currents into any junction ...

$$\sum I = 0 \quad \text{... equals zero.} \quad (26.5)$$

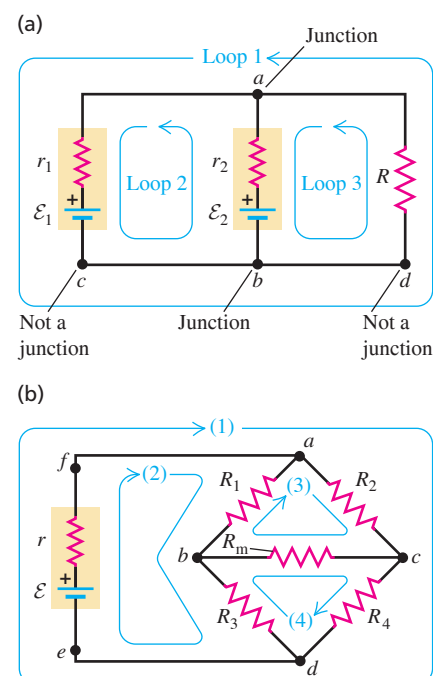
Kirchhoff's loop rule
(valid for any closed loop):

The sum of the potential differences around any loop ...

$$\sum V = 0 \quad \text{... equals zero.} \quad (26.6)$$

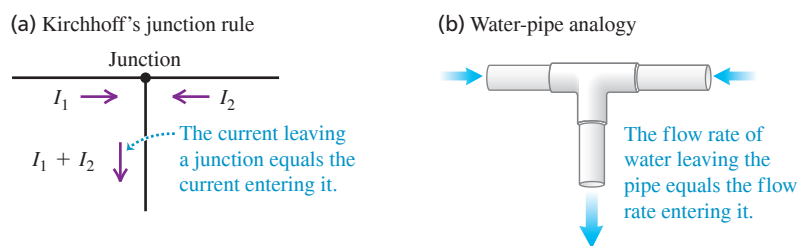
Note that the potential differences V in Eq. (26.6) include those associated with all circuit elements in the loop, including emfs and resistors.

Figure 26.6 Two networks that cannot be reduced to simple series-parallel combinations of resistors.



CAUTION **Current need not split equally at a junction** As a rule, at a junction where the current in a wire splits to follow two or more separate paths, the current does *not* split equally among the different paths. More current flows along a path that has less resistance. ■

Figure 26.7 Kirchhoff's junction rule states that as much current flows into a junction as flows out of it.



The junction rule is based on *conservation of electric charge*. No charge can accumulate at a junction, so the total charge entering the junction per unit time must equal the total charge leaving per unit time (**Fig. 26.7a**). Charge per unit time is current, so if we consider the currents entering a junction to be positive and those leaving to be negative, the algebraic sum of currents into a junction must be zero. It's like a T branch in a water pipe (**Fig. 26.7b**); if you have a total of 1 liter per minute coming in the two pipes, you can't have 3 liters per minute going out the third pipe. We used the junction rule (without saying so) in Section 26.1 in the derivation of Eq. (26.2) for resistors in parallel.

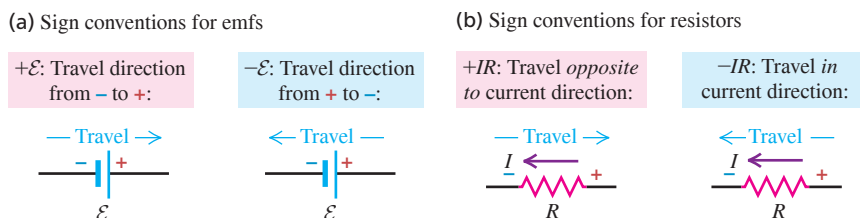
The loop rule is a statement that the electrostatic force is *conservative*. Suppose we go around a loop, measuring potential differences across successive circuit elements as we go. When we return to the starting point, we must find that the *algebraic sum* of these differences is zero; otherwise, we could not say that the potential at this point has a definite value.

Sign Conventions for the Loop Rule

In applying the loop rule, we need some sign conventions. Problem-Solving Strategy 26.2 describes in detail how to use these, but here's a quick overview. We first assume a direction for the current in each branch of the circuit and mark it on a diagram of the circuit. Then, starting at any point in the circuit, we imagine traveling around a loop, adding emfs and IR terms as we come to them. When we travel through a source in the direction from $-$ to $+$, the emf is considered to be *positive*; when we travel from $+$ to $-$, the emf is considered to be *negative* (**Fig. 26.8a**). When we travel through a resistor in the *same* direction as the assumed current, the IR term is *negative* because the current goes in the direction of decreasing potential. When we travel through a resistor in the direction *opposite* to the assumed current, the IR term is *positive* because this represents a rise of potential (**Fig. 26.8b**).

Kirchhoff's two rules are all we need to solve a wide variety of network problems. Usually, some of the emfs, currents, and resistances are known, and others are unknown. We must always obtain from Kirchhoff's rules a number of independent equations equal to the number of unknowns so that we can solve the equations simultaneously. Often the hardest part of the solution is keeping track of algebraic signs!

Figure 26.8 Use these sign conventions when you apply Kirchhoff's loop rule. In each part of the figure "Travel" is the direction that we imagine going around the loop, which is not necessarily the direction of the current.



PROBLEM-SOLVING STRATEGY 26.2 Kirchhoff's Rules

IDENTIFY the relevant concepts: Kirchhoff's rules are useful for analyzing any electric circuit.

SET UP the problem using the following steps:

1. Draw a circuit diagram, leaving room to label all quantities, known and unknown. Indicate an assumed direction for each unknown current and emf. (Kirchhoff's rules will yield the magnitudes and directions of unknown currents and emfs. If the actual direction of a quantity is opposite to your assumption, the resulting quantity will have a negative sign.)
2. As you label currents, it's helpful to use Kirchhoff's junction rule, as in Fig. 26.9, so as to express the currents in terms of as few quantities as possible.
3. Identify the target variables.

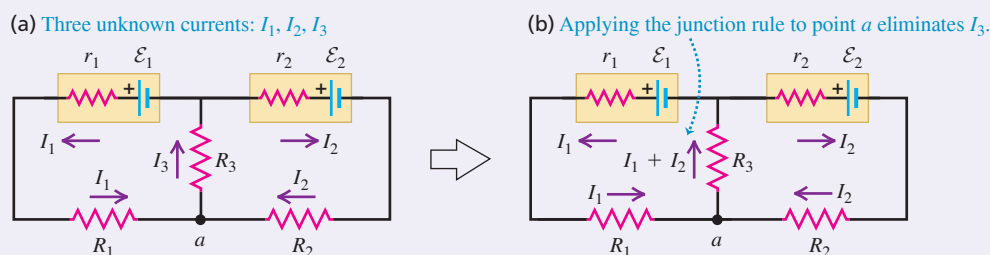
EXECUTE the solution as follows:

1. Choose any loop in the network and choose a direction (clockwise or counterclockwise) to travel around the loop as you apply Kirchhoff's loop rule. The direction need not be the same as any assumed current direction.

2. Travel around the loop in the chosen direction, adding potential differences algebraically as you cross them. Use the sign conventions of Fig. 26.8.
3. Equate the sum obtained in step 2 to zero in accordance with the loop rule.
4. If you need more independent equations, choose another loop and repeat steps 1–3; continue until you have as many independent equations as unknowns or until every circuit element has been included in at least one loop.
5. Solve the equations simultaneously to determine the unknowns.
6. You can use the loop-rule bookkeeping system to find the potential V_{ab} of any point a with respect to any other point b . Start at b and add the potential changes you encounter in going from b to a ; use the same sign rules as in step 2. The algebraic sum of these changes is $V_{ab} = V_a - V_b$.

EVALUATE your answer: Check all the steps in your algebra. Apply steps 1 and 2 to a loop you have not yet considered; if the sum of potential drops isn't zero, you've made an error somewhere.

Figure 26.9 Applying the junction rule to point a reduces the number of unknown currents from three to two.



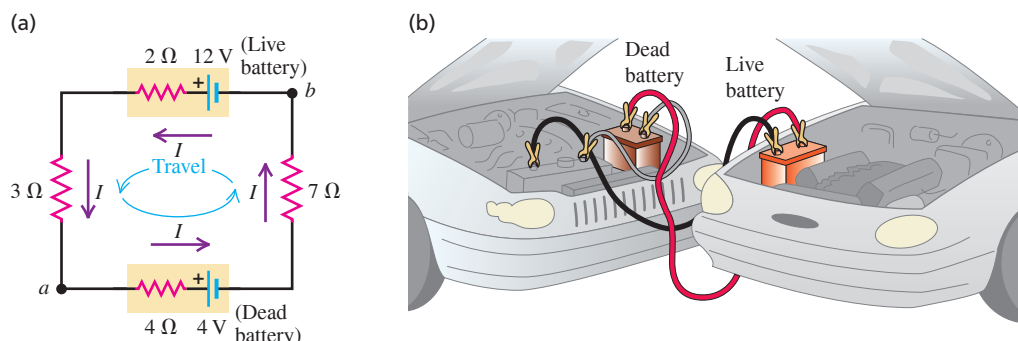
EXAMPLE 26.3 A single-loop circuit

WITH VARIATION PROBLEMS

The circuit shown in Fig. 26.10a contains two batteries, each with an emf and an internal resistance, and two resistors. Find (a) the current in the circuit, (b) the potential difference V_{ab} , and (c) the power output of the emf of each battery.

IDENTIFY and SET UP There are no junctions in this single-loop circuit, so we don't need Kirchhoff's junction rule. To apply Kirchhoff's loop rule, we first assume a direction for the current; let's assume a counterclockwise direction as shown in Fig. 26.10a.

Figure 26.10 (a) In this example we travel around the loop in the same direction as the assumed current, so all the IR terms are negative. The potential decreases as we travel from $+$ to $-$ through the bottom emf but increases as we travel from $-$ to $+$ through the top emf. (b) A real-life example of a circuit of this kind.



Continued

EXECUTE (a) Starting at a and traveling counterclockwise with the current, we add potential increases and decreases and equate the sum to zero as in Eq. (26.6):

$$-I(4\ \Omega) - 4\text{ V} - I(7\ \Omega) + 12\text{ V} - I(2\ \Omega) - I(3\ \Omega) = 0$$

Collecting like terms and solving for I , we find

$$8\text{ V} = I(16\ \Omega) \quad \text{and} \quad I = 0.5\text{ A}$$

The positive result for I shows that our assumed current direction is correct.

(b) To find V_{ab} , the potential at a with respect to b , we start at b and add potential changes as we go toward a . There are two paths from b to a ; taking the lower one, we find

$$V_{ab} = (0.5\text{ A})(7\ \Omega) + 4\text{ V} + (0.5\text{ A})(4\ \Omega) = 9.5\text{ V}$$

Point a is at 9.5 V higher potential than b . All the terms in this sum, including the IR terms, are positive because each represents an *increase* in potential as we go from b to a . For the upper path,

$$V_{ab} = 12\text{ V} - (0.5\text{ A})(2\ \Omega) - (0.5\text{ A})(3\ \Omega) = 9.5\text{ V}$$

Here the IR terms are negative because our path goes in the direction of the current, with potential decreases through the resistors. The results for V_{ab} are the same for both paths, as they must be in order for the total potential change around the loop to be zero.

(c) The power outputs of the emf of the 12 V and 4 V batteries are

$$P_{12\text{V}} = \mathcal{E}I = (12\text{ V})(0.5\text{ A}) = 6\text{ W}$$

$$P_{4\text{V}} = \mathcal{E}I = (-4\text{ V})(0.5\text{ A}) = -2\text{ W}$$

The negative sign in $\mathcal{E}$ for the 4 V battery appears because the current actually runs from the higher-potential side of the battery to the lower-potential side. The negative value of P means that we are *storing* energy in that battery; the 12 V battery is *recharging* it (if it is in fact rechargeable; otherwise, we're destroying it).

EVALUATE By applying the expression $P = I^2R$ to each of the four resistors in Fig. 26.10a, you can show that the total power dissipated in all four resistors is 4 W. Of the 6 W provided by the emf of the 12 V battery, 2 W goes into storing energy in the 4 V battery and 4 W is dissipated in the resistances.

The circuit shown in Fig. 26.10a is much like that used when a fully charged 12 V storage battery (in a car with its engine running) “jump-starts” a car with a run-down battery (Fig. 26.10b). The run-down battery is slightly recharged in the process. The 3 Ω and 7 Ω resistors in Fig. 26.10a represent the resistances of the jumper cables and of the conducting path through the car with the run-down battery. (The values of the resistances in actual automobiles and jumper cables are considerably lower, and the emf of a run-down car battery isn't much less than 12 V.)

KEYCONCEPT In any loop of a circuit, Kirchhoff's loop rule applies: The sum of the potential changes as you travel around the loop must be zero. If the circuit has only a single loop, the current has the same value everywhere in the circuit.

EXAMPLE 26.4 Charging a battery

WITH VARIATION PROBLEMS

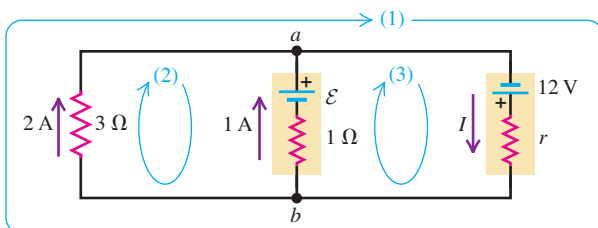
In the circuit shown in **Fig. 26.11**, a 12 V power supply with unknown internal resistance r is connected to a run-down rechargeable battery with unknown emf $\mathcal{E}$ and internal resistance 1 Ω and to an indicator light bulb of resistance 3 Ω carrying a current of 2 A. The current through the run-down battery is 1 A in the direction shown. Find r , $\mathcal{E}$, and the current I through the power supply.

IDENTIFY and SET UP This circuit has more than one loop, so we must apply both the junction and loop rules. We assume the direction of the current through the 12 V power supply, and the polarity of the run-down battery, to be as shown in Fig. 26.11. There are three target variables, so we need three equations.

EXECUTE We apply the junction rule, Eq. (26.5), to point a :

$$-I + 1\text{ A} + 2\text{ A} = 0 \quad \text{so} \quad I = 3\text{ A}$$

Figure 26.11 In this circuit a power supply charges a run-down battery and lights a bulb. An assumption has been made about the polarity of the emf $\mathcal{E}$ of the battery. Is this assumption correct?



To determine r , we apply the loop rule, Eq. (26.6), to the large, outer loop (1):

$$12\text{ V} - (3\text{ A})r - (2\text{ A})(3\ \Omega) = 0 \quad \text{so} \quad r = 2\ \Omega$$

To determine $\mathcal{E}$, we apply the loop rule to the left-hand loop (2):

$$-\mathcal{E} + (1\text{ A})(1\ \Omega) - (2\text{ A})(3\ \Omega) = 0 \quad \text{so} \quad \mathcal{E} = -5\text{ V}$$

The negative value for $\mathcal{E}$ shows that the actual polarity of this emf is opposite to that shown in Fig. 26.11. As in Example 26.3, the battery is being recharged.

EVALUATE Try applying the junction rule at point b instead of point a , and try applying the loop rule counterclockwise rather than clockwise around loop (1). You'll get the same results for I and r . We can check our result for $\mathcal{E}$ by using loop (3):

$$12\text{ V} - (3\text{ A})(2\ \Omega) - (1\text{ A})(1\ \Omega) + \mathcal{E} = 0$$

which again gives us $\mathcal{E} = -5\text{ V}$.

As an additional check, we note that $V_{ba} = V_b - V_a$ equals the voltage across the 3 Ω resistance, which is $(2\text{ A})(3\ \Omega) = 6\text{ V}$. Going from a to b by the right-hand branch, we encounter potential differences $+12\text{ V} - (3\text{ A})(2\ \Omega) = +6\text{ V}$, and going by the middle branch, we find $-(-5\text{ V}) + (1\text{ A})(1\ \Omega) = +6\text{ V}$. The three ways of getting V_{ba} give the same results.

KEYCONCEPT In any circuit that has more than one loop, Kirchhoff's junction rule applies: At each junction, the sum of the currents into the junction must be zero.

EXAMPLE 26.5 Power in a battery-charging circuit**WITH VARIATION PROBLEMS**

In the circuit of Example 26.4 (shown in Fig. 26.11), find the power delivered by the 12 V power supply and by the battery being recharged, and find the power dissipated in each resistor.

IDENTIFY and SET UP We use the results of Section 25.5, in which we found that the power delivered *from* an emf to a circuit is $\mathcal{E}I$ and the power delivered *to* a resistor from a circuit is $V_{ab}I = I^2R$. We know the values of all relevant quantities from Example 26.4.

EXECUTE The power output from the emf of the power supply is

$$P_{\text{supply}} = \mathcal{E}_{\text{supply}}I_{\text{supply}} = (12 \text{ V})(3 \text{ A}) = 36 \text{ W}$$

The power dissipated in the power supply's internal resistance r is

$$P_{r\text{-supply}} = I_{\text{supply}}^2 r_{\text{supply}} = (3 \text{ A})^2(2 \Omega) = 18 \text{ W}$$

so the power supply's *net* power output is $P_{\text{net}} = 36 \text{ W} - 18 \text{ W} = 18 \text{ W}$. Alternatively, from Example 26.4 the terminal voltage of the battery is $V_{ba} = 6 \text{ V}$, so the net power output is

$$P_{\text{net}} = V_{ba}I_{\text{supply}} = (6 \text{ V})(3 \text{ A}) = 18 \text{ W}$$

The power output of the emf $\mathcal{E}$ of the battery being charged is

$$P_{\text{emf}} = \mathcal{E}I_{\text{battery}} = (-5 \text{ V})(1 \text{ A}) = -5 \text{ W}$$

This is negative because the 1 A current runs through the battery from the higher-potential side to the lower-potential side. (As we mentioned in Example 26.4, the polarity assumed for this battery in Fig. 26.11 was wrong.) We are storing energy in the battery as we charge it. Additional power is dissipated in the battery's internal resistance; this power is

$$P_{r\text{-battery}} = I_{\text{battery}}^2 r_{\text{battery}} = (1 \text{ A})^2(1 \Omega) = 1 \text{ W}$$

The total power input to the battery is thus $1 \text{ W} + |-5 \text{ W}| = 6 \text{ W}$. Of this, 5 W represents useful energy stored in the battery; the remainder is wasted in its internal resistance. The power dissipated in the light bulb is

$$P_{\text{bulb}} = I_{\text{bulb}}^2 R_{\text{bulb}} = (2 \text{ A})^2(3 \Omega) = 12 \text{ W}$$

EVALUATE As a check, note that all of the power from the supply is accounted for. Of the 18 W of net power from the power supply, 5 W goes to recharge the battery, 1 W is dissipated in the battery's internal resistance, and 12 W is dissipated in the light bulb.

KEYCONCEPT In any circuit that contains a source (such as a battery or power supply) and resistors, the sum of the power dissipated in each resistor must equal the power supplied by the source.

EXAMPLE 26.6 A complex network**WITH VARIATION PROBLEMS**

Figure 26.12 shows a “bridge” circuit of the type described at the beginning of this section (see Fig. 26.6b). Find the current in each resistor and the equivalent resistance of the network of five resistors.

IDENTIFY and SET UP This network is neither a series combination nor a parallel combination. Hence we must use Kirchhoff's rules to find the values of the target variables. There are five unknown currents, but by applying the junction rule to junctions a and b , we can represent them in terms of three unknown currents I_1 , I_2 , and I_3 , as shown in Fig. 26.12.

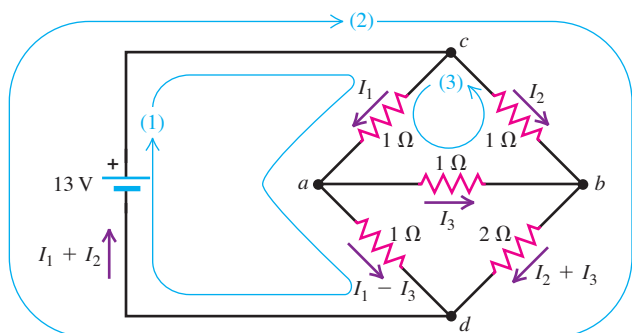
EXECUTE We apply the loop rule to the three loops shown:

$$13 \text{ V} - I_1(1 \Omega) - (I_1 - I_3)(1 \Omega) = 0 \quad (1)$$

$$-I_2(1 \Omega) - (I_2 + I_3)(2 \Omega) + 13 \text{ V} = 0 \quad (2)$$

$$-I_1(1 \Omega) - I_3(1 \Omega) + I_2(1 \Omega) = 0 \quad (3)$$

Figure 26.12 A network circuit with several resistors.



One way to solve these simultaneous equations is to solve Eq. (3) for I_2 , obtaining $I_2 = I_1 + I_3$, and then substitute this expression into Eq. (2) to eliminate I_2 . We then have

$$13 \text{ V} = I_1(2 \Omega) - I_3(1 \Omega) \quad (1')$$

$$13 \text{ V} = I_1(3 \Omega) + I_3(5 \Omega) \quad (2')$$

Now we can eliminate I_3 by multiplying Eq. (1') by 5 and adding the two equations. We obtain

$$78 \text{ V} = I_1(13 \Omega) \quad I_1 = 6 \text{ A}$$

We substitute this result into Eq. (1') to obtain $I_3 = -1 \text{ A}$, and from Eq. (3) we find $I_2 = 5 \text{ A}$. The negative value of I_3 tells us that its direction is opposite to the direction we assumed.

The total current through the network is $I_1 + I_2 = 11 \text{ A}$, and the potential drop across it is equal to the battery emf, 13 V. The equivalent resistance of the network is therefore

$$R_{\text{eq}} = \frac{13 \text{ V}}{11 \text{ A}} = 1.2 \Omega$$

EVALUATE You can check our results for I_1 , I_2 , and I_3 by substituting them back into Eqs. (1)–(3). What do you find?

KEYCONCEPT Some circuits have combinations of resistors that are neither in series nor in parallel. To analyze any such circuit, use Kirchhoff's loop rule and junction rule.

EXAMPLE 26.7 A potential difference in a complex network

WITH VARIATION PROBLEMS

In the circuit of Example 26.6 (Fig. 26.12), find the potential difference V_{ab} .

IDENTIFY and SET UP Our target variable $V_{ab} = V_a - V_b$ is the potential at point a with respect to point b . To find it, we start at point b and follow a path to point a , adding potential rises and drops as we go. We can follow any of several paths from b to a ; the result must be the same for all such paths, which gives us a way to check our result.

EXECUTE The simplest path is through the center $1\ \Omega$ resistor. In Example 26.6 we found $I_3 = -1\ \text{A}$, showing that the actual current direction through this resistor is from right to left. Thus, as we go from b to a , there is a *drop* of potential with magnitude $|I_3|R = (1\ \text{A})(1\ \Omega) = 1\ \text{V}$. Hence $V_{ab} = -1\ \text{V}$, and the potential at a is $1\ \text{V}$ less than at point b .

EVALUATE To check our result, let's try a path from b to a that goes through the lower two resistors. The currents through these are

$$I_2 + I_3 = 5\ \text{A} + (-1\ \text{A}) = 4\ \text{A} \quad \text{and}$$

$$I_1 - I_3 = 6\ \text{A} - (-1\ \text{A}) = 7\ \text{A}$$

and so

$$V_{ab} = -(4\ \text{A})(2\ \Omega) + (7\ \text{A})(1\ \Omega) = -1\ \text{V}$$

You can confirm this result by using some other paths from b to a .

KEYCONCEPT A useful way to restate Kirchhoff's loop rule is that for any path in a circuit from point a to point b , the potential difference V_{ab} must be the same.

TEST YOUR UNDERSTANDING OF SECTION 26.2 Subtract Eq. (1) from Eq. (2) in Example 26.6. To which loop in Fig. 26.12 does this equation correspond? Would this equation have simplified the solution of Example 26.6?

ANSWER

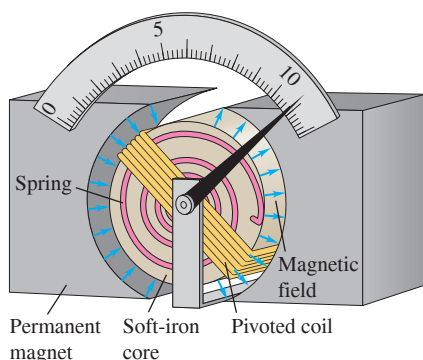
with the solution of Example 26.6. $I_1 - I_3 = 6\ \text{A} - (-1\ \text{A}) = 7\ \text{A}$. We can obtain this equation by applying the loop rule around the path from c to b to d to a to c in Fig. 26.12. This isn't an independent equation, so it would not have helped.

Figure 26.13 Both this voltmeter (left) and ammeter (right) are d'Arsonval galvanometers. The difference has to do with their internal connections (see Fig. 26.15).



Figure 26.14 A d'Arsonval galvanometer, showing a pivoted coil with attached pointer, a permanent magnet supplying a magnetic field that is uniform in magnitude, and a spring to provide restoring torque, which opposes magnetic-field torque.

Magnetic-field torque tends to push pointer away from zero. Spring torque tends to push pointer toward zero.



26.3 ELECTRICAL MEASURING INSTRUMENTS

We've been talking about potential difference, current, and resistance for two chapters, so it's about time we said something about how to *measure* these quantities. Many common devices, including car instrument panels, battery chargers, and inexpensive electrical instruments, measure potential difference (voltage), current, or resistance with a **d'Arsonval galvanometer** (Fig. 26.13). In the following discussion we'll often call it just a *meter*. A pivoted coil of fine wire is placed in the magnetic field of a permanent magnet (Fig. 26.14). Attached to the coil is a spring, similar to the hairspring on the balance wheel of a watch. In the equilibrium position, with no current in the coil, the pointer is at zero. When there is a current in the coil, the magnetic field exerts a torque on the coil that is proportional to the current. (We'll discuss this magnetic interaction in detail in Chapter 27.) As the coil turns, the spring exerts a restoring torque that is proportional to the angular displacement.

Thus the angular deflection of the coil and pointer is directly proportional to the coil current, and the device can be calibrated to measure current. The maximum deflection, typically 90° or so, is called *full-scale deflection*. The essential electrical characteristics of the meter are the current I_{fs} required for full-scale deflection (typically on the order of $10\ \mu\text{A}$ to $10\ \text{mA}$) and the resistance R_c of the coil (typically on the order of $10\ \Omega$ to $1000\ \Omega$).

The meter deflection is proportional to the *current* in the coil. If the coil obeys Ohm's law, the current is proportional to the *potential difference* between the terminals of the coil, and the deflection is also proportional to this potential difference. For example, consider a meter whose coil has a resistance $R_c = 20.0\ \Omega$ and that deflects full scale when the current in its coil is $I_{fs} = 1.00\ \text{mA}$. The corresponding potential difference for full-scale deflection is

$$V = I_{fs}R_c = (1.00 \times 10^{-3}\ \text{A})(20.0\ \Omega) = 0.0200\ \text{V}$$

Ammeters

A current-measuring instrument is usually called an **ammeter** (or milliammeter, microammeter, and so forth, depending on the range). An *ammeter always measures the current passing through it*. An *ideal* ammeter, discussed in Section 25.4, would have *zero* resistance, so including it in a branch of a circuit would not affect the current in that branch.

Real ammeters always have a finite resistance, but it is always desirable for an ammeter to have as little resistance as possible.

We can adapt any meter to measure currents that are larger than its full-scale reading by connecting a resistor in parallel with it (Fig. 26.15a) so that some of the current bypasses the meter coil. The parallel resistor is called a **shunt resistor** or simply a *shunt*, denoted as R_{sh} .

Suppose we want to make a meter with full-scale current I_{fs} and coil resistance R_c into an ammeter with full-scale reading I_a . To determine the shunt resistance R_{sh} needed, note that at full-scale deflection the total current through the parallel combination is I_a , the current through the coil of the meter is I_{fs} , and the current through the shunt is the difference $I_a - I_{\text{fs}}$. The potential difference V_{ab} is the same for both paths, so

$$I_{\text{fs}}R_c = (I_a - I_{\text{fs}})R_{\text{sh}} \quad (\text{for an ammeter}) \quad (26.7)$$

EXAMPLE 26.8 Designing an ammeter

What shunt resistance is required to make the 1.00 mA, 20.0 Ω meter described above into an ammeter with a range of 0 to 50.0 mA?

IDENTIFY and SET UP Since the meter is being used as an ammeter, its internal connections are as shown in Fig. 26.15a. Our target variable is the shunt resistance R_{sh} , which we'll find from Eq. (26.7). The ammeter must handle a maximum current $I_a = 50.0 \times 10^{-3}$ A. The coil resistance is $R_c = 20.0 \Omega$, and the meter shows full-scale deflection when the current through the coil is $I_{\text{fs}} = 1.00 \times 10^{-3}$ A.

EXECUTE Solving Eq. (26.7) for R_{sh} , we find

$$\begin{aligned} R_{\text{sh}} &= \frac{I_{\text{fs}}R_c}{I_a - I_{\text{fs}}} = \frac{(1.00 \times 10^{-3} \text{ A})(20.0 \Omega)}{(50.0 \times 10^{-3} \text{ A}) - (1.00 \times 10^{-3} \text{ A})} \\ &= 0.408 \Omega \end{aligned}$$

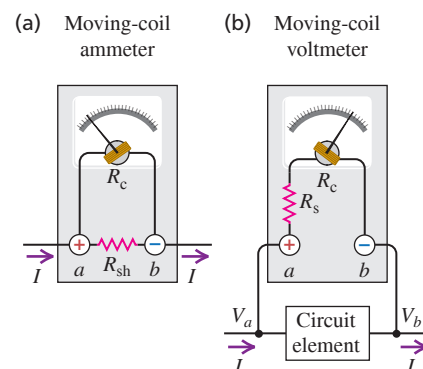
EVALUATE It's useful to consider the equivalent resistance R_{eq} of the ammeter as a whole. From Eq. (26.2),

$$\begin{aligned} R_{\text{eq}} &= \left(\frac{1}{R_c} + \frac{1}{R_{\text{sh}}} \right)^{-1} = \left(\frac{1}{20.0 \Omega} + \frac{1}{0.408 \Omega} \right)^{-1} \\ &= 0.400 \Omega \end{aligned}$$

The shunt resistance is so small in comparison to the coil resistance that the equivalent resistance is very nearly equal to the shunt resistance. The result is an ammeter with a low equivalent resistance and the desired 0–50.0 mA range. At full-scale deflection, $I = I_a = 50.0$ mA, the current through the galvanometer is 1.00 mA, the current through the shunt resistor is 49.0 mA, and $V_{ab} = 0.0200$ V. If the current I is less than 50.0 mA, the coil current and the deflection are proportionally less.

KEYCONCEPT An ammeter measures the current that passes through it. It should have a very small equivalent resistance in order to minimize its effect on the circuit being measured. This can be accomplished using a small shunt resistance in parallel with the meter.

Figure 26.15 Using the same meter to measure (a) current and (b) voltage.



Voltmeters

This same basic meter may also be used to measure potential difference or *voltage*. A voltage-measuring device is called a **voltmeter**. A voltmeter always measures the potential difference between two points, and its terminals must be connected to these points. (Example 25.6 in Section 25.4 described what can happen if a voltmeter is connected incorrectly.) As we discussed in Section 25.4, an ideal voltmeter would have *infinite* resistance, so connecting it between two points in a circuit would not alter any of the currents. Real voltmeters always have finite resistance, but a voltmeter should have large enough resistance that connecting it in a circuit does not change the other currents appreciably.

For the meter of Example 26.8, the voltage across the meter coil at full-scale deflection is only $I_{\text{fs}}R_c = (1.00 \times 10^{-3} \text{ A})(20.0 \Omega) = 0.0200$ V. We can extend this range by connecting a resistor R_s in *series* with the coil (Fig. 26.15b). Then only a fraction of the total potential difference appears across the coil itself, and the remainder appears across R_s . For a voltmeter with full-scale reading V_V , we need a series resistor R_s in Fig. 26.15b such that

$$V_V = I_{\text{fs}}(R_c + R_s) \quad (\text{for a voltmeter}) \quad (26.8)$$

BIO APPLICATION

Electromyography A fine needle containing two electrodes is being inserted into a muscle in this patient's hand. By using a sensitive voltmeter to measure the potential difference between these electrodes, a physician can probe the muscle's electrical activity. This is an important technique for diagnosing neurological and neuromuscular diseases.



EXAMPLE 26.9 Designing a voltmeter

What series resistance is required to make the 1.00 mA, 20.0 Ω meter described above into a voltmeter with a range of 0 to 10.0 V?

IDENTIFY and SET UP Since this meter is being used as a voltmeter, its internal connections are as shown in Fig. 26.15b. The maximum allowable voltage across the voltmeter is $V_V = 10.0$ V. We want this to occur when the current through the coil is $I_{fs} = 1.00 \times 10^{-3}$ A. Our target variable is the series resistance R_s , which we find from Eq. (26.8).

EXECUTE From Eq. (26.8),

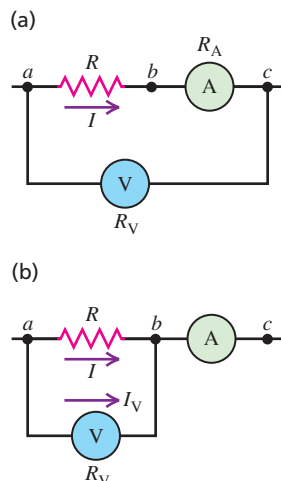
$$R_s = \frac{V_V}{I_{fs}} - R_c = \frac{10.0 \text{ V}}{0.00100 \text{ A}} - 20.0 \Omega = 9980 \Omega$$

EVALUATE At full-scale deflection, $V_{ab} = 10.0$ V, the voltage across the meter is 0.0200 V, the voltage across R_s is 9.98 V, and the current

through the voltmeter is 0.00100 A. Most of the voltage appears across the series resistor. The meter's equivalent resistance is a desirably high $R_{eq} = 20.0 \Omega + 9980 \Omega = 10,000 \Omega$. Such a meter is called a "1000 ohms-per-volt" meter, referring to the ratio of resistance to full-scale deflection. In normal operation the current through the circuit element being measured (I in Fig. 26.15b) is much greater than 0.00100 A, and the resistance between points a and b in the circuit is much less than 10,000 Ω . The voltmeter draws off only a small fraction of the current and thus disturbs the circuit being measured only slightly.

KEYCONCEPT A voltmeter measures the potential difference (voltage) between its terminals. It should have a very large equivalent resistance in order to minimize the amount of current it draws from the circuit being measured. This can be accomplished using a large shunt resistance in series with the meter.

Figure 26.16 Ammeter–voltmeter method for measuring resistance.

**Ammeters and Voltmeters in Combination**

A voltmeter and an ammeter can be used together to measure *resistance* and *power*. The resistance R of a resistor equals the potential difference V_{ab} between its terminals divided by the current I ; that is, $R = V_{ab}/I$. The power input P to any circuit element is the product of the potential difference across it and the current through it: $P = V_{ab}I$. In principle, the most straightforward way to measure R or P is to measure V_{ab} and I simultaneously.

With practical ammeters and voltmeters this isn't quite as simple as it seems. In **Fig. 26.16a**, ammeter A reads the current I in the resistor R . Voltmeter V , however, reads the *sum* of the potential difference V_{ab} across the resistor and the potential difference V_{bc} across the ammeter. If we transfer the voltmeter terminal from c to b , as in **Fig. 26.16b**, then the voltmeter reads the potential difference V_{ab} correctly, but the ammeter now reads the *sum* of the current I in the resistor and the current I_V in the voltmeter. Either way, we have to correct the reading of one instrument or the other unless the corrections are small enough to be negligible.

EXAMPLE 26.10 Measuring resistance I

The voltmeter in the circuit of Fig. 26.16a reads 12.0 V and the ammeter reads 0.100 A. The meter resistances are $R_V = 10,000 \Omega$ (for the voltmeter) and $R_A = 2.00 \Omega$ (for the ammeter). What are the resistance R and the power dissipated in the resistor?

IDENTIFY and SET UP The ammeter reads the current $I = 0.100$ A through the resistor, and the voltmeter reads the potential difference between a and c . If the ammeter were *ideal* (that is, if $R_A = 0$), there would be zero potential difference between b and c , the voltmeter reading $V = 12.0$ V would be equal to the potential difference V_{ab} across the resistor, and the resistance would be equal to $R = V/I = (12.0 \text{ V})/(0.100 \text{ A}) = 120 \Omega$. The ammeter is *not* ideal, however (its resistance is $R_A = 2.00 \Omega$), so the voltmeter reading V is actually the sum of the potential differences V_{bc} (across the ammeter) and V_{ab} (across the resistor). We use Ohm's law to find the voltage V_{bc} from the known current and ammeter resistance. Then we solve for V_{ab} and R . Given these, we are able to calculate the power P into the resistor.

EXECUTE From Ohm's law, $V_{bc} = IR_A = (0.100 \text{ A})(2.00 \Omega) = 0.200 \text{ V}$ and $V_{ab} = IR$. The sum of these is $V = 12.0 \text{ V}$, so the potential difference

across the resistor is $V_{ab} = V - V_{bc} = (12.0 \text{ V}) - (0.200 \text{ V}) = 11.8 \text{ V}$. Hence the resistance is

$$R = \frac{V_{ab}}{I} = \frac{11.8 \text{ V}}{0.100 \text{ A}} = 118 \Omega$$

The power dissipated in this resistor is

$$P = V_{ab}I = (11.8 \text{ V})(0.100 \text{ A}) = 1.18 \text{ W}$$

EVALUATE You can confirm this result for the power by using the alternative formula $P = I^2R$. Do you get the same answer?

KEYCONCEPT To find the resistance of a resistor and the power that the resistor dissipates requires measuring both the current through the resistor and the potential difference across the resistor. To do this, put an ammeter in series with the resistance and put a voltmeter across the resistor and ammeter.

EXAMPLE 26.11 Measuring resistance II

Suppose the meters of Example 26.10 are connected to a different resistor as shown in Fig. 26.16b, and the readings obtained on the meters are the same as in Example 26.10. What is the value of this new resistance R , and what is the power dissipated in the resistor?

IDENTIFY and SET UP In Example 26.10 the ammeter read the actual current through the resistor, but the voltmeter reading was not the same as the potential difference across the resistor. Now the situation is reversed: The voltmeter reading $V = 12.0$ V shows the actual potential difference V_{ab} across the resistor, but the ammeter reading $I_A = 0.100$ A is *not* equal to the current I through the resistor. Applying the junction rule at b in Fig. 26.16b shows that $I_A = I + I_V$, where I_V is the current through the voltmeter. We find I_V from the given values of V and the voltmeter resistance R_V , and we use this value to find the resistor current I . We then determine the resistance R from I and the voltmeter reading, and calculate the power as in Example 26.10.

EXECUTE We have $I_V = V/R_V = (12.0 \text{ V})/(10,000 \Omega) = 1.20 \text{ mA}$. The actual current I in the resistor is $I = I_A - I_V = 0.100 \text{ A} - 0.0012 \text{ A} = 0.0988 \text{ A}$, and the resistance is

$$R = \frac{V_{ab}}{I} = \frac{12.0 \text{ V}}{0.0988 \text{ A}} = 121 \Omega$$

The power dissipated in the resistor is

$$P = V_{ab}I = (12.0 \text{ V})(0.0988 \text{ A}) = 1.19 \text{ W}$$

EVALUATE Had the meters been ideal, our results would have been $R = 12.0 \text{ V}/0.100 \text{ A} = 120 \Omega$ and $P = VI = (12.0 \text{ V})(0.100 \text{ A}) = 1.2 \text{ W}$ both here and in Example 26.10. The actual (correct) results are not too different in either case. That's because the ammeter and voltmeter are nearly ideal: Compared with the resistance R under test, the ammeter resistance R_A is very small and the voltmeter resistance R_V is very large. Under these conditions, treating the meters as ideal yields pretty good results; accurate work requires calculations as in these two examples.

KEYCONCEPT An alternative way to find the resistance of a resistor and the power that the resistor dissipates is to put a voltmeter across the resistor and put an ammeter in the circuit downstream of the resistor–voltmeter combination.

Ohmmeters

An alternative method for measuring resistance is to use a d'Arsonval meter in an arrangement called an **ohmmeter**. It consists of a meter, a resistor, and a source (often a flashlight battery) connected in series (Fig. 26.17). The resistance R to be measured is connected between terminals x and y .

The series resistance R_s is variable; it is adjusted so that when terminals x and y are short-circuited (that is, when $R = 0$), the meter deflects full scale. When nothing is connected to terminals x and y , so that the circuit between x and y is *open* (that is, when $R \rightarrow \infty$), there is no current and hence no deflection. For any intermediate value of R the meter deflection depends on the value of R , and the meter scale can be calibrated to read the resistance R directly. Larger currents correspond to smaller resistances, so this scale reads backward compared to the scale showing the current.

In situations in which high precision is required, instruments containing d'Arsonval meters have been supplanted by electronic instruments with direct digital readouts. Digital voltmeters can be made with extremely high internal resistance, of the order of $100 \text{ M}\Omega$. **Figure 26.18** shows a digital *multimeter*, an instrument that can measure voltage, current, or resistance over a wide range.

Figure 26.17 Ohmmeter circuit. The resistor R_s has a variable resistance, as is indicated by the arrow through the resistor symbol. To use the ohmmeter, first connect x directly to y and adjust R_s until the meter reads zero. Then connect x and y across the resistor R and read the scale.

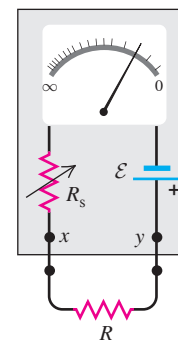


Figure 26.18 This digital multimeter can be used as a voltmeter (red arc), ammeter (yellow arc), or ohmmeter (green arc).



The Potentiometer

The *potentiometer* is an instrument that can be used to measure the emf of a source without drawing any current from the source; it also has a number of other useful applications. Essentially, it balances an unknown potential difference against an adjustable, measurable potential difference.

Figure 26.19a shows the principle of the potentiometer. A resistance wire ab of total resistance R_{ab} is permanently connected to the terminals of a source of known emf $\mathcal{E}_1$. A sliding contact c is connected through the galvanometer G to a second source whose emf $\mathcal{E}_2$ is to be measured. As contact c is moved along the resistance wire, the resistance R_{cb} between points c and b varies; if the resistance wire is uniform, R_{cb} is proportional to the length of wire between c and b . To determine the value of $\mathcal{E}_2$, contact c is moved until a position is found at which the galvanometer shows no deflection; this corresponds to zero current passing through $\mathcal{E}_2$. With $I_2 = 0$, Kirchhoff's loop rule gives

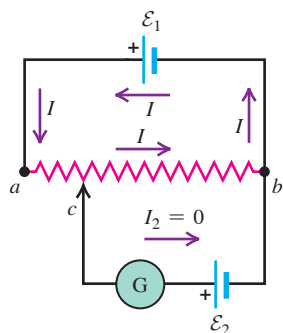
$$\mathcal{E}_2 = IR_{cb}$$

With $I_2 = 0$, the current I produced by the emf $\mathcal{E}_1$ has the same value no matter what the value of the emf $\mathcal{E}_2$. We calibrate the device by replacing $\mathcal{E}_2$ by a source of known emf; then to find any unknown emf $\mathcal{E}_2$, we measure the length of wire cb for which $I_2 = 0$. Note: For this to work, V_{ab} must be greater than $\mathcal{E}_2$.

The term *potentiometer* is also used for any variable resistor, usually having a circular resistance element and a sliding contact controlled by a rotating shaft and knob. Figure 26.19b shows the circuit symbol for a potentiometer.

Figure 26.19 A potentiometer.

(a) Potentiometer circuit



(b) Circuit symbol for potentiometer (variable resistor)



TEST YOUR UNDERSTANDING OF SECTION 26.3 You want to measure the current through and the potential difference across the $2\ \Omega$ resistor shown in Fig. 26.12 (Example 26.6 in Section 26.2). (a) How should you connect an ammeter and a voltmeter to do this? (i) Both ammeter and voltmeter in series with the $2\ \Omega$ resistor; (ii) ammeter in series with the $2\ \Omega$ resistor and voltmeter connected between points b and d ; (iii) ammeter connected between points b and d and voltmeter in series with the $2\ \Omega$ resistor; (iv) both ammeter and voltmeter connected between points b and d . (b) What resistances should these meters have? (i) Both ammeter and voltmeter resistances should be much greater than $2\ \Omega$; (ii) ammeter resistance should be much greater than $2\ \Omega$ and voltmeter resistance should be much less than $2\ \Omega$; (iii) ammeter resistance should be much less than $2\ \Omega$ and voltmeter resistance should be much greater than $2\ \Omega$; (iv) both ammeter and voltmeter resistances should be much less than $2\ \Omega$.

ANSWER

(a) (i) An ammeter must always be placed in series with the circuit element of interest, and a voltmeter must always be placed in parallel. Ideally the ammeter would have zero resistance and the voltmeter would have infinite resistance so that their presence would have no effect on either the resistor current or the voltage. Neither of these idealizations is possible, but the ammeter resistance should be much less than $2\ \Omega$ and the voltmeter resistance should be much greater than $2\ \Omega$.

26.4 R-C CIRCUITS

In the circuits we have analyzed up to this point, we have assumed that all the emfs and resistances are *constant* (time independent) so that all the potentials, currents, and powers are also independent of time. But in the simple act of charging or discharging a capacitor we find a situation in which the currents, voltages, and powers *do* change with time.

Many devices incorporate circuits in which a capacitor is alternately charged and discharged. These include flashing traffic lights, automobile turn signals, and electronic flash units. Understanding what happens in such circuits is thus of great practical importance.

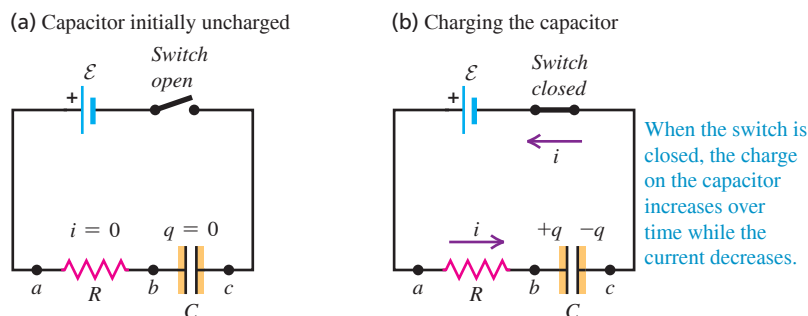


Figure 26.20 Charging a capacitor. (a) Just before the switch is closed, the charge q is zero. (b) When the switch closes (at $t = 0$), the current jumps from zero to $\mathcal{E}/R$. As time passes, q approaches Q_f and the current i approaches zero.

Charging a Capacitor

Figure 26.20 shows a simple circuit for charging a capacitor. A circuit such as this that has a resistor and a capacitor in series is called an **R-C circuit**. We idealize the battery (or power supply) to have a constant emf $\mathcal{E}$ and zero internal resistance ($r = 0$), and we ignore the resistance of all the connecting conductors.

We begin with the capacitor initially uncharged (Fig. 26.20a); then at some initial time $t = 0$ we close the switch, completing the circuit and permitting current around the loop to begin charging the capacitor (Fig. 26.20b). For all practical purposes, the current begins at the same instant in every conducting part of the circuit, and at each instant the current is the same in every part.

Because the capacitor in Fig. 26.20 is initially uncharged, the potential difference v_{bc} across it is zero at $t = 0$. At this time, from Kirchhoff's loop law, the voltage v_{ab} across the resistor R is equal to the battery emf $\mathcal{E}$. The initial ($t = 0$) current through the resistor, which we'll call I_0 , is given by Ohm's law: $I_0 = v_{ab}/R = \mathcal{E}/R$.

As the capacitor charges, its voltage v_{bc} increases and the potential difference v_{ab} across the resistor decreases, corresponding to a decrease in current. The sum of these two voltages is constant and equal to $\mathcal{E}$. After a long time the capacitor is fully charged, the current decreases to zero, and v_{ab} across the resistor becomes zero. Then the entire battery emf $\mathcal{E}$ appears across the capacitor and $v_{bc} = \mathcal{E}$.

Let q represent the charge on the capacitor and i the current in the circuit at some time t after the switch has been closed. We choose the positive direction for the current to correspond to positive charge flowing onto the left-hand capacitor plate, as in Fig. 26.20b. The instantaneous potential differences v_{ab} and v_{bc} are

$$v_{ab} = iR \quad v_{bc} = \frac{q}{C}$$

Using these in Kirchhoff's loop rule, we find

$$\mathcal{E} - iR - \frac{q}{C} = 0 \quad (26.9)$$

The potential drops by an amount iR as we travel from a to b and by q/C as we travel from b to c . Solving Eq. (26.9) for i , we find

$$i = \frac{\mathcal{E}}{R} - \frac{q}{RC} \quad (26.10)$$

At time $t = 0$, when the switch is first closed, the capacitor is uncharged, and so $q = 0$. Substituting $q = 0$ into Eq. (26.10), we find that the *initial* current I_0 is given by $I_0 = \mathcal{E}/R$, as we have already noted. If the capacitor were not in the circuit, the last term in Eq. (26.10) would not be present; then the current would be *constant* and equal to $\mathcal{E}/R$.

As the charge q increases, the term q/RC becomes larger and the capacitor charge approaches its final value, which we'll call Q_f . The current decreases and eventually becomes zero. When $i = 0$, Eq. (26.10) gives

$$\frac{\mathcal{E}}{R} = \frac{Q_f}{RC} \quad Q_f = C\mathcal{E} \quad (26.11)$$

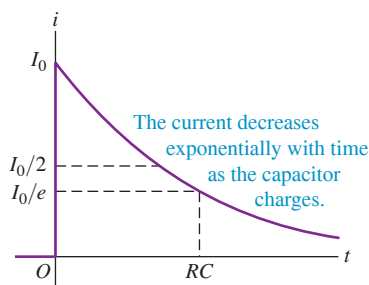
Note that the final charge Q_f does not depend on R .

CAUTION Lowercase means time-varying

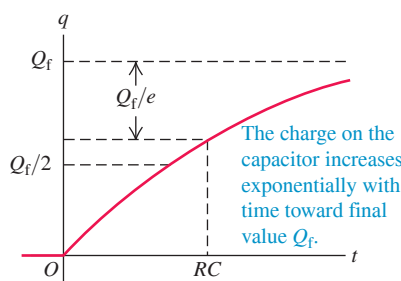
Up to this point we have been working with constant potential differences (voltages), currents, and charges, and we have used *capital* letters V , I , and Q , respectively, to denote these quantities. To distinguish between quantities that vary with time and those that are constant, we'll use *lowercase* letters v , i , and q for time-varying voltages, currents, and charges, respectively. We suggest that you follow this same convention in your own work. ■

Figure 26.21 Current i and capacitor charge q as functions of time for the circuit of Fig. 26.20. The initial current is I_0 and the initial capacitor charge is zero. The current asymptotically approaches zero, and the capacitor charge asymptotically approaches a final value of Q_f .

(a) Graph of current versus time for a charging capacitor



(b) Graph of capacitor charge versus time for a charging capacitor



BIO APPLICATION Pacemakers and Capacitors

This x-ray image shows a pacemaker implanted in a patient with a malfunctioning sinoatrial node, the part of the heart that generates the electrical signal to trigger heartbeats. The pacemaker circuit contains a battery, a capacitor, and a computer-controlled switch. To maintain regular beating, once per second the switch discharges the capacitor and sends an electrical pulse along the lead to the heart. The switch then flips to allow the capacitor to recharge for the next pulse.

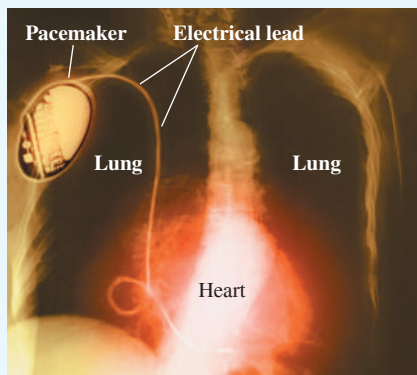


Figure 26.21 shows the current and capacitor charge as functions of time. At the instant the switch is closed ($t = 0$), the current jumps from zero to its initial value $I_0 = \mathcal{E}/R$; after that, it gradually approaches zero. The capacitor charge starts at zero and gradually approaches the final value given by Eq. (26.11), $Q_f = C\mathcal{E}$.

We can derive general expressions for charge q and current i as functions of time. With our choice of the positive direction for current (Fig. 26.20b), i equals the rate at which positive charge arrives at the left-hand (positive) plate of the capacitor, so $i = dq/dt$. Making this substitution in Eq. (26.10), we have

$$\frac{dq}{dt} = \frac{\mathcal{E}}{R} - \frac{q}{RC} = -\frac{1}{RC}(q - C\mathcal{E})$$

We can rearrange this to

$$\frac{dq}{q - C\mathcal{E}} = -\frac{dt}{RC}$$

and then integrate both sides. We change the integration variables to q' and t' so that we can use q and t for the upper limits. The lower limits are $q' = 0$ and $t' = 0$:

$$\int_0^q \frac{dq'}{q' - C\mathcal{E}} = -\int_0^t \frac{dt'}{RC}$$

When we carry out the integration, we get

$$\ln\left(\frac{q - C\mathcal{E}}{-C\mathcal{E}}\right) = -\frac{t}{RC}$$

Exponentiating both sides (that is, taking the inverse logarithm) and solving for q , we find

$$\frac{q - C\mathcal{E}}{-C\mathcal{E}} = e^{-t/RC}$$

R-C circuit, charging capacitor:

$$q = C\mathcal{E}(1 - e^{-t/RC}) = Q_f(1 - e^{-t/RC}) \quad (26.12)$$

Labels in diagram: Capacitor charge, Capacitance, Final capacitor charge = $C\mathcal{E}$, Battery emf, Time since switch closed, Resistance.

The instantaneous current i is just the time derivative of Eq. (26.12):

R-C circuit, charging capacitor:

$$i = \frac{dq}{dt} = \frac{\mathcal{E}}{R} e^{-t/RC} = I_0 e^{-t/RC} = \mathcal{E}/R \quad (26.13)$$

Labels in diagram: Current, Battery emf, Time since switch closed, Rate of change of capacitor charge, Resistance, Capacitance, Initial current = $\mathcal{E}/R$.

The charge and current are both *exponential* functions of time. Figure 26.21a is a graph of Eq. (26.13) and Fig. 26.21b is a graph of Eq. (26.12).

Time Constant

After a time equal to RC , the current in the R - C circuit has decreased to $1/e$ (about 0.368) of its initial value. At this time, the capacitor charge has reached $(1 - 1/e) = 0.632$ of its final value $Q_f = C\mathcal{E}$. The product RC is therefore a measure of how quickly the capacitor charges. We call RC the **time constant**, or the **relaxation time**, of the circuit, denoted by τ :

$$\tau = RC \quad (\text{time constant for } R\text{-}C \text{ circuit}) \quad (26.14)$$

When τ is small, the capacitor charges quickly; when it is larger, the charging takes more time. If the resistance is small, it's easier for current to flow, and the capacitor charges more quickly. If R is in ohms and C in farads, τ is in seconds.

In Fig. 26.21a the horizontal axis is an *asymptote* for the curve. Strictly speaking, i never becomes exactly zero. But the longer we wait, the closer it gets. After a time equal to $10RC$, the current has decreased to 0.000045 of its initial value. Similarly, the curve in Fig. 26.21b approaches the horizontal dashed line labeled Q_f as an asymptote. The charge q never attains exactly this value, but after a time equal to $10RC$, the difference between q and Q_f is only 0.000045 of Q_f . We invite you to verify that the product RC has units of time.

Discharging a Capacitor

Now suppose that after the capacitor in Fig. 26.21b has acquired a charge Q_0 , we remove the battery from our R - C circuit and connect points a and c to an open switch (Fig. 26.22a). We then close the switch and at the same instant reset our stopwatch to $t = 0$; at that time, $q = Q_0$. The capacitor then *discharges* through the resistor, and its charge eventually decreases to zero.

Again let i and q represent the time-varying current and charge at some instant after the connection is made. In Fig. 26.22b we make the same choice of the positive direction for current as in Fig. 26.20b. Then Kirchhoff's loop rule gives Eq. (26.10) but with $\mathcal{E} = 0$; that is,

$$i = \frac{dq}{dt} = -\frac{q}{RC} \quad (26.15)$$

The current i is now negative; this is because positive charge q is leaving the left-hand capacitor plate in Fig. 26.22b, so the current is in the direction opposite to that shown. At time $t = 0$, when $q = Q_0$, the initial current is $I_0 = -Q_0/RC$.

To find q as a function of time, we rearrange Eq. (26.15), again change the variables to q' and t' , and integrate. This time the limits for q' are Q_0 to q :

$$\int_{Q_0}^q \frac{dq'}{q'} = -\frac{1}{RC} \int_0^t dt'$$

$$\ln \frac{q}{Q_0} = -\frac{t}{RC}$$

R-C circuit, discharging capacitor:

$$q = Q_0 e^{-t/RC} \quad (26.16)$$

Labels in diagram: Capacitor charge, Initial capacitor charge, Capacitance, Resistance, Time since switch closed.

The instantaneous current i is the derivative of this with respect to time:

R-C circuit, discharging capacitor:

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC} e^{-t/RC} = I_0 e^{-t/RC} \quad (26.17)$$

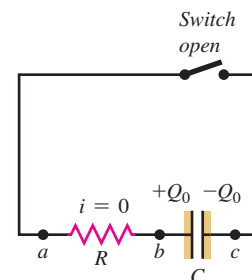
Labels in diagram: Current, Initial capacitor charge, Capacitance, Resistance, Time since switch closed, Rate of change of capacitor charge, Initial current = $-Q_0/RC$.

We graph the current and the charge in Fig. 26.23; both quantities approach zero exponentially with time. Comparing these results with Eqs. (26.12) and (26.13), we note that the expressions for the current are identical, apart from the sign of I_0 . The capacitor charge approaches zero asymptotically in Eq. (26.16), while the *difference* between q and Q approaches zero asymptotically in Eq. (26.12).

Energy considerations give us additional insight into the behavior of an R - C circuit. While the capacitor is charging, the instantaneous rate at which the battery delivers energy to the circuit is $P = \mathcal{E}i$. The instantaneous rate at which electrical energy is dissipated in

Figure 26.22 Discharging a capacitor. (a) Before the switch is closed at time $t = 0$, the capacitor charge is Q_0 and the current is zero. (b) At time t after the switch is closed, the capacitor charge is q and the current is i . The actual current direction is opposite to the direction shown; i is negative. After a long time, q and i both approach zero.

(a) Capacitor initially charged



(b) Discharging the capacitor

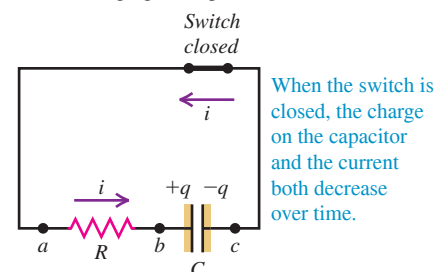
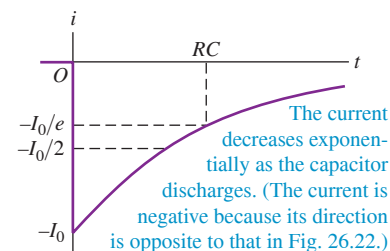
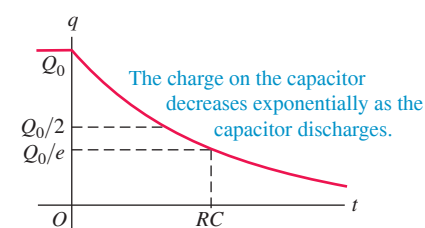


Figure 26.23 Current i and capacitor charge q as functions of time for the circuit of Fig. 26.22. The initial current is I_0 and the initial capacitor charge is Q_0 . Both i and q asymptotically approach zero.

(a) Graph of current versus time for a discharging capacitor



(b) Graph of capacitor charge versus time for a discharging capacitor



the resistor is i^2R , and the rate at which energy is stored in the capacitor is $iv_{bc} = iq/C$. Multiplying Eq. (26.9) by i , we find

$$\mathcal{E}i = i^2R + \frac{iq}{C} \quad (26.18)$$

This means that of the power $\mathcal{E}i$ supplied by the battery, part (i^2R) is dissipated in the resistor and part (iq/C) is stored in the capacitor.

The *total* energy supplied by the battery during charging of the capacitor equals the battery emf $\mathcal{E}$ multiplied by the total charge Q_f , or $\mathcal{E}Q_f$. The total energy stored in the capacitor, from Eq. (24.9), is $Q_f\mathcal{E}/2$. Thus, of the energy supplied by the battery, *exactly half* is stored in the capacitor, and the other half is dissipated in the resistor. This half-and-half division of energy doesn't depend on C , R , or $\mathcal{E}$. You can verify this result by taking the integral over time of each of the power quantities in Eq. (26.18).

EXAMPLE 26.12 Charging a capacitor

WITH VARIATION PROBLEMS

A $10\text{ M}\Omega$ resistor is connected in series with a $1.0\text{ }\mu\text{F}$ capacitor and a battery with emf 12.0 V . Before the switch is closed at time $t = 0$, the capacitor is uncharged. (a) What is the time constant? (b) What fraction of the final charge Q_f is on the capacitor at $t = 46\text{ s}$? (c) What fraction of the initial current I_0 is still flowing at $t = 46\text{ s}$?

IDENTIFY and SET UP This is the situation shown in Fig. 26.20, with $R = 10\text{ M}\Omega$, $C = 1.0\text{ }\mu\text{F}$, and $\mathcal{E} = 12.0\text{ V}$. The charge q and current i vary with time as shown in Fig. 26.21. Our target variables are (a) the time constant τ , (b) the ratio q/Q_f at $t = 46\text{ s}$, and (c) the ratio i/I_0 at $t = 46\text{ s}$. Equation (26.14) gives τ . For a capacitor being charged, Eq. (26.12) gives q and Eq. (26.13) gives i .

EXECUTE (a) From Eq. (26.14),

$$\tau = RC = (10 \times 10^6\text{ }\Omega)(1.0 \times 10^{-6}\text{ F}) = 10\text{ s}$$

(b) From Eq. (26.12),

$$\frac{q}{Q_f} = 1 - e^{-t/RC} = 1 - e^{-(46\text{ s})/(10\text{ s})} = 0.99$$

(c) From Eq. (26.13),

$$\frac{i}{I_0} = e^{-t/RC} = e^{-(46\text{ s})/(10\text{ s})} = 0.010$$

EVALUATE After 4.6 time constants the capacitor is 99% charged and the charging current has decreased to 1.0% of its initial value. The circuit would charge more rapidly if we reduced the time constant by using a smaller resistance.

KEYCONCEPT When a capacitor is charging in a circuit with a resistor and a source of emf, the capacitor charge q and the current i both vary with time: q approaches its final value asymptotically and i approaches zero asymptotically. The time constant for both quantities is the product RC of the resistance and the capacitance of the circuit elements.

EXAMPLE 26.13 Discharging a capacitor

WITH VARIATION PROBLEMS

The resistor and capacitor of Example 26.12 are reconnected as shown in Fig. 26.22. The capacitor has an initial charge of $5.0\text{ }\mu\text{C}$ and is discharged by closing the switch at $t = 0$. (a) At what time will the charge be $0.50\text{ }\mu\text{C}$? (b) What is the current at this time?

IDENTIFY and SET UP Now the capacitor is being discharged, so q and i vary with time as in Fig. 26.23, with $Q_0 = 5.0 \times 10^{-6}\text{ C}$. Again we have $RC = \tau = 10\text{ s}$. Our target variables are (a) the value of t at which $q = 0.50\text{ }\mu\text{C}$ and (b) the value of i at this time. We first solve Eq. (26.16) for t , and then solve Eq. (26.17) for i .

EXECUTE (a) Solving Eq. (26.16) for the time t gives

$$\begin{aligned} t &= -RC \ln \frac{q}{Q_0} = -(10\text{ s}) \ln \frac{0.50\text{ }\mu\text{C}}{5.0\text{ }\mu\text{C}} \\ &= 23\text{ s} = 2.3\tau \end{aligned}$$

(b) From Eq. (26.17), with $Q_0 = 5.0\text{ }\mu\text{C} = 5.0 \times 10^{-6}\text{ C}$,

$$i = -\frac{Q_0}{RC} e^{-t/RC} = -\frac{5.0 \times 10^{-6}\text{ C}}{10\text{ s}} e^{-2.3} = -5.0 \times 10^{-8}\text{ A}$$

EVALUATE The current in part (b) is negative because i has the opposite sign when the capacitor is discharging than when it is charging. Note that we could have avoided evaluating $e^{-t/RC}$ by noticing that at the time in question, $q = 0.10Q_0$; from Eq. (26.16) this means that $e^{-t/RC} = 0.10$.

KEYCONCEPT When a capacitor is discharging in a circuit with a resistor, the capacitor charge q and the current i both vary with time, and both approach zero asymptotically. The time constant for both quantities is the product RC of the resistance and the capacitance of the circuit elements.

TEST YOUR UNDERSTANDING OF SECTION 26.4 The energy stored in a capacitor is equal to $q^2/2C$. When a capacitor is discharged, what fraction of the initial energy remains after an elapsed time of one time constant? (i) $1/e$; (ii) $1/e^2$; (iii) $1 - 1/e$; (iv) $(1 - 1/e)^2$; (v) answer depends on how much energy was stored initially.

ANSWER

(ii) After one time constant, $t = RC$ and the initial charge Q_0 has decreased to $Q_0 e^{-t/RC} = Q_0 e^{-1} = Q_0/e$. Hence the stored energy has decreased from $Q_0^2/2C$ to $(Q_0/e)^2/2C = Q_0^2/2C e^2$, a fraction $1/e^2 = 0.135$ of its initial value. This result doesn't depend on the initial value of the energy.

26.5 POWER DISTRIBUTION SYSTEMS

We conclude this chapter with a brief discussion of practical household and automotive electric-power distribution systems. Automobiles use direct-current (dc) systems, while nearly all household, commercial, and industrial systems use alternating current (ac) because of the ease of stepping voltage up and down with transformers. Most of the same basic wiring concepts apply to both. We'll talk about alternating-current circuits in greater detail in Chapter 31.

The various lamps, motors, and other appliances to be operated are always connected in *parallel* to the power source (the wires from the power company for houses, or from the battery and alternator for a car). If appliances were connected in series, shutting one appliance off would shut them all off (see Example 26.2 in Section 26.1). **Figure 26.24** shows the basic idea of house wiring. One side of the “line,” as the pair of conductors is called, is called the *neutral* side; it is always connected to “ground” at the entrance panel. For houses, *ground* is an actual electrode driven into the earth (which is usually a good conductor) or sometimes connected to the household water pipes. Electricians speak of the “hot” side and the “neutral” side of the line. Most modern house wiring systems have *two* hot lines with opposite polarity with respect to the neutral. We'll return to this detail later.

Household voltage is nominally 120 V in the United States and Canada, and often 240 V in Europe. (For alternating current, which varies sinusoidally with time, these numbers represent the *root-mean-square* voltage, which is $1/\sqrt{2}$ times the peak voltage. We'll discuss this further in Section 31.1.) The amount of current I drawn by a given device is determined by its power input P , given by Eq. (25.17): $P = VI$. Hence $I = P/V$. For example, the current in a 100 W light bulb is

$$I = \frac{P}{V} = \frac{100 \text{ W}}{120 \text{ V}} = 0.83 \text{ A}$$

The power input to this bulb is actually determined by its resistance R . Using Eq. (25.18), which states that $P = VI = I^2 R = V^2/R$ for a resistor, we get the resistance of this bulb at operating temperature:

$$R = \frac{V}{I} = \frac{120 \text{ V}}{0.83 \text{ A}} = 144 \, \Omega \quad \text{or} \quad R = \frac{V^2}{P} = \frac{(120 \text{ V})^2}{100 \text{ W}} = 144 \, \Omega$$

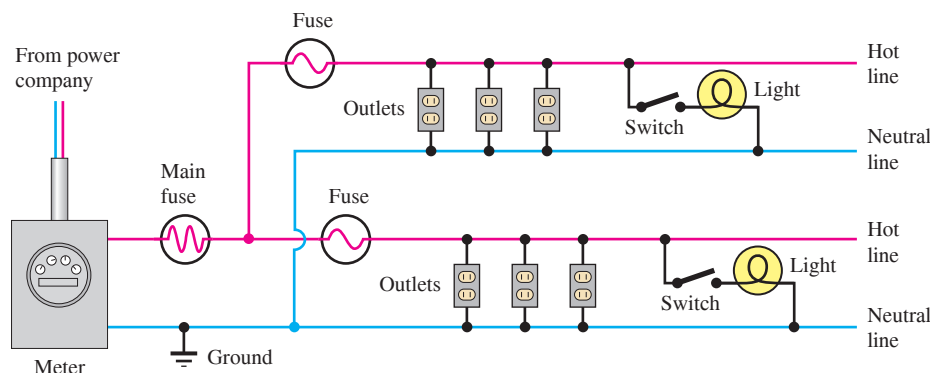


Figure 26.24 Schematic diagram of part of a house wiring system. Only two branch circuits are shown; an actual system might have four to thirty branch circuits. Lamps and appliances may be plugged into the outlets. The grounding wires, which normally carry no current, are not shown.

Similarly, a 1500 W waffle iron draws a current of $(1500 \text{ W})/(120 \text{ V}) = 12.5 \text{ A}$ and has a resistance, at operating temperature, of 9.6Ω . Because of the temperature dependence of resistivity, the resistances of these devices are considerably less when they are cold. If you measure the resistance of a 100 W light bulb with an ohmmeter (whose small current causes very little temperature rise), you'll probably get a value of about 10Ω . When a light bulb is turned on, this low resistance causes an initial surge of current until the filament heats up. That's why a light bulb that's ready to burn out nearly always does so just when you turn it on.

Circuit Overloads and Short Circuits

The maximum current available from an individual circuit is limited by the resistance of the wires. As we discussed in Section 25.5, the I^2R power loss in the wires causes them to become hot, and in extreme cases this can cause a fire or melt the wires. Ordinary lighting and outlet wiring in houses usually uses 12 gauge wire. This has a diameter of 2.05 mm and can carry a maximum current of 20 A safely (without overheating). Larger-diameter wires of the same length have lower resistance [see Eq. (25.10)]. Hence 8 gauge (3.26 mm) or 6 gauge (4.11 mm) is used for high-current appliances such as clothes dryers, and 2 gauge (6.54 mm) or larger is used for the main power lines entering a house.

Protection against overloading and overheating of circuits is provided by fuses or circuit breakers. A *fuse* contains a link of lead–tin alloy with a very low melting temperature; the link melts and breaks the circuit when its rated current is exceeded (**Fig. 26.25a**). A *circuit breaker* is an electromechanical device that performs the same function, using an electromagnet or a bimetallic strip to “trip” the breaker and interrupt the circuit when the current exceeds a specified value (**Fig. 26.25b**). Circuit breakers have the advantage that they can be reset after they are tripped, while a blown fuse must be replaced.

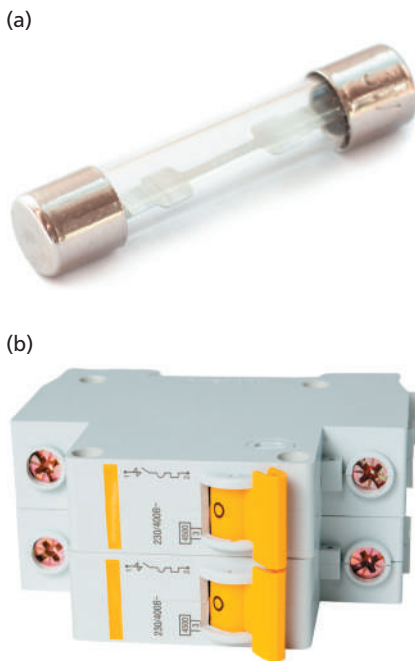
CAUTION Fuses If your system has fuses and you plug too many high-current appliances into the same outlet, the fuse blows. Do not replace the fuse with one that has a higher rating; if you do, you risk overheating the wires and starting a fire. The only safe solution is to distribute the appliances among several circuits. Modern kitchens often have three or four separate 20 A circuits. ■

Contact between the hot and neutral sides of the line causes a *short circuit*. Such a situation, which can be caused by faulty insulation or by a variety of mechanical malfunctions, provides a very low-resistance current path, permitting a very large current that would quickly melt the wires and ignite their insulation if the current were not interrupted by a fuse or circuit breaker (see Example 25.10 in Section 25.5). An equally dangerous situation is a broken wire that interrupts the current path, creating an *open circuit*. This is hazardous because of the sparking that can occur at the point of intermittent contact.

In approved wiring practice, a fuse or breaker is placed *only* in the hot side of the line, never in the neutral side. Otherwise, if a short circuit should develop because of faulty insulation or other malfunction, the ground-side fuse could blow. The hot side would still be live and would pose a shock hazard if you touched the live conductor and a grounded object such as a water pipe. For similar reasons the wall switch for a light fixture should always be in the hot side of the line, never the neutral side.

Further protection against shock hazard is provided by a third conductor called the *grounding wire*, included in all present-day wiring. This conductor corresponds to the long round or U-shaped prong of the three-prong connector plug on an appliance or power tool. It is connected to the neutral side of the line at the entrance panel. The grounding wire normally carries no current, but it connects the metal case or frame of the device to ground. If a conductor on the hot side of the line accidentally contacts the frame or case, the grounding conductor provides a current path, and the fuse blows. Without the ground wire, the frame could become “live”—that is, at a potential 120 V above ground. Then if you touched it and a water pipe (or even a damp floor) at the same time, you could get a

Figure 26.25 (a) Excess current will melt the thin wire of lead–tin alloy that runs along the length of a fuse, inside the transparent housing. (b) The switch on this circuit breaker will flip if the maximum allowable current is exceeded.



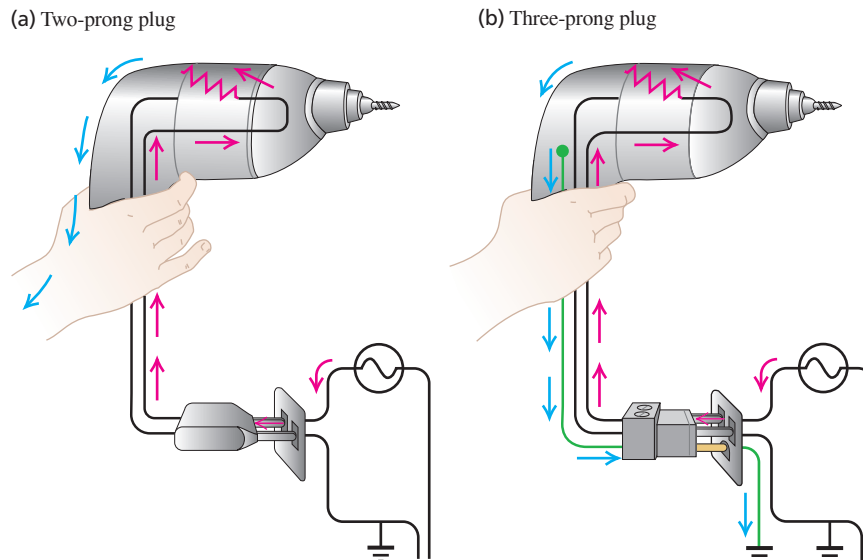


Figure 26.26 (a) If a malfunctioning electric drill is connected to a wall socket via a two-prong plug, a person may receive a shock. (b) When the drill malfunctions when connected via a three-prong plug, a person touching it receives no shock, because electric charge flows through the ground wire (shown in green) to the third prong and into the ground rather than into the person's body. If the ground current is appreciable, the fuse blows.

dangerous shock (**Fig. 26.26**). In some situations, especially outlets located outdoors or near a sink or other water pipes, a special kind of circuit breaker called a *ground-fault interrupter* (GFI or GFCI) is used. This device senses the difference in current between the hot and neutral conductors (which is normally zero) and trips when some very small value, typically 5 mA, is exceeded.

Household and Automotive Wiring

Most modern household wiring systems actually use a slight elaboration of the system described above. The power company provides *three* conductors. One is neutral; the other two are both at 120 V with respect to the neutral but with opposite polarity, giving a voltage between them of 240 V. The power company calls this a *three-wire line*, in contrast to the 120 V two-wire (plus ground wire) line described above. With a three-wire line, 120 V lamps and appliances can be connected between neutral and either hot conductor, and high-power devices requiring 240 V, such as electric ranges and clothes dryers, are connected between the two hot lines.

All of the above discussion can be applied directly to automobile wiring. The voltage is about 13 V (direct current); the power is supplied by the battery and by the alternator, which charges the battery when the engine is running. The neutral side of each circuit is connected to the body and frame of the vehicle. For this low voltage a separate grounding conductor is not required for safety. The fuse or circuit breaker arrangement is the same in principle as in household wiring. Because of the lower voltage (less energy per charge), more current (a greater number of charges per second) is required for the same power; a 100 W headlight bulb requires a current of about $(100 \text{ W})/(13 \text{ V}) = 8 \text{ A}$.

Although we spoke of *power* in the above discussion, what we buy from the power company is *energy*. Power is energy transferred per unit time, so energy is average power multiplied by time. The usual unit of energy sold by the power company is the kilowatt-hour (1 kW · h):

$$1 \text{ kW} \cdot \text{h} = (10^3 \text{ W})(3600 \text{ s}) = 3.6 \times 10^6 \text{ W} \cdot \text{s} = 3.6 \times 10^6 \text{ J}$$

In the United States, one kilowatt-hour typically costs 8 to 27 cents, depending on the location and quantity of energy purchased. To operate a 1500 W (1.5 kW) waffle iron continuously for 1 hour requires 1.5 kW · h of energy; at 10 cents per kilowatt-hour, the energy cost is 15 cents. The cost of operating any lamp or appliance for a specified time can be calculated in the same way if the power rating is known. However, many electric cooking utensils (including waffle irons) cycle on and off to maintain a constant temperature, so the average power may be less than the power rating marked on the device.

EXAMPLE 26.14 A kitchen circuit

An 1800 W toaster, a 1.3 kW electric frying pan, and a 100 W lamp are plugged into the same 20 A, 120 V circuit. (a) What current is drawn by each device, and what is the resistance of each device? (b) Will this combination trip the circuit breaker?

IDENTIFY and SET UP When plugged into the same circuit, the three devices are connected in parallel, so the voltage across each appliance is $V = 120$ V. We find the current I drawn by each device from the relationship $P = VI$, where P is the power input of the device. To find the resistance R of each device we use the relationship $P = V^2/R$.

EXECUTE (a) To simplify the calculation of current and resistance, we note that $I = P/V$ and $R = V^2/P$. Hence

$$\begin{aligned} I_{\text{toaster}} &= \frac{1800 \text{ W}}{120 \text{ V}} = 15 \text{ A} & R_{\text{toaster}} &= \frac{(120 \text{ V})^2}{1800 \text{ W}} = 8 \Omega \\ I_{\text{frying pan}} &= \frac{1300 \text{ W}}{120 \text{ V}} = 11 \text{ A} & R_{\text{frying pan}} &= \frac{(120 \text{ V})^2}{1300 \text{ W}} = 11 \Omega \\ I_{\text{lamp}} &= \frac{100 \text{ W}}{120 \text{ V}} = 0.83 \text{ A} & R_{\text{lamp}} &= \frac{(120 \text{ V})^2}{100 \text{ W}} = 144 \Omega \end{aligned}$$

For constant voltage the device with the *least* resistance (in this case the toaster) draws the most current and receives the most power.

(b) The total current through the line is the sum of the currents drawn by the three devices:

$$\begin{aligned} I &= I_{\text{toaster}} + I_{\text{frying pan}} + I_{\text{lamp}} \\ &= 15 \text{ A} + 11 \text{ A} + 0.83 \text{ A} = 27 \text{ A} \end{aligned}$$

This exceeds the 20 A rating of the line, and the circuit breaker will indeed trip.

EVALUATE We could also find the total current by using $I = P/V$ and the total power P delivered to all three devices:

$$\begin{aligned} I &= \frac{P_{\text{toaster}} + P_{\text{frying pan}} + P_{\text{lamp}}}{V} \\ &= \frac{1800 \text{ W} + 1300 \text{ W} + 100 \text{ W}}{120 \text{ V}} = 27 \text{ A} \end{aligned}$$

A third way to determine I is to use $I = V/R_{\text{eq}}$, where R_{eq} is the equivalent resistance of the three devices in parallel:

$$I = \frac{V}{R_{\text{eq}}} = (120 \text{ V}) \left(\frac{1}{8 \Omega} + \frac{1}{11 \Omega} + \frac{1}{144 \Omega} \right) = 27 \text{ A}$$

Appliances with such current demands are common, so modern kitchens have more than one 20 A circuit. To keep currents safely below 20 A, the toaster and frying pan should be plugged into different circuits.

KEYCONCEPT The voltage across any device plugged into a household wiring system has the same value. The smaller the resistance of the device, the more current it draws and the more power it receives.

TEST YOUR UNDERSTANDING OF SECTION 26.5 To prevent the circuit breaker in Example 26.14 from blowing, a home electrician replaces the circuit breaker with one rated at 40 A. Is this a reasonable thing to do?

ANSWER

no This is a very dangerous thing to do. The circuit breaker will allow currents up to 40 A, double the rated value of the wiring. The amount of power $P = I^2 R$ dissipated in a section of wire can therefore be up to four times the rated value, so the wires could get hot and start a fire. (This assumes the resistance R remains unchanged. In fact, R increases with temperature, so the dissipated power can be even greater, than we have estimated.)

CHAPTER 26 SUMMARY

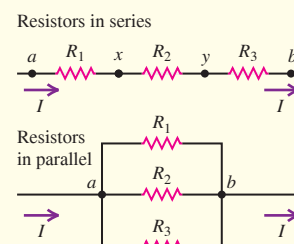
Resistors in series and parallel: When several resistors $R_1, R_2, R_3, \dots$ are connected in series, the equivalent resistance R_{eq} is the sum of the individual resistances. The same *current* flows through all the resistors in a series connection. When several resistors are connected in parallel, the reciprocal of equivalent resistance R_{eq} is the sum of the reciprocals of the individual resistances. All resistors in a parallel connection have the same *potential difference* between their terminals. (See Examples 26.1 and 26.2.)

$$R_{\text{eq}} = R_1 + R_2 + R_3 + \dots \quad (26.1)$$

(resistors in series)

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \quad (26.2)$$

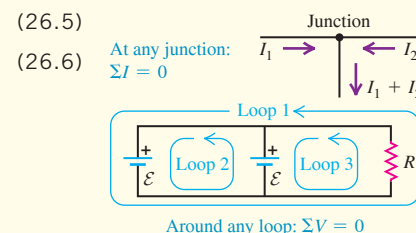
(resistors in parallel)



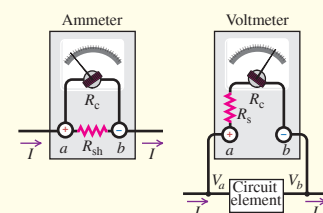
Kirchhoff's rules: Kirchhoff's junction rule is based on conservation of charge. It states that the algebraic sum of the currents into any junction must be zero. Kirchhoff's loop rule is based on conservation of energy and the conservative nature of electrostatic fields. It states that the algebraic sum of potential differences around any loop must be zero. Careful use of consistent sign rules is essential in applying Kirchhoff's rules. (See Examples 26.3–26.7.)

$$\sum I = 0 \quad (\text{junction rule})$$

$$\sum V = 0 \quad (\text{loop rule})$$



Electrical measuring instruments: In a d'Arsonval galvanometer, the deflection is proportional to the current in the coil. For a larger current range, a shunt resistor is added, so some of the current bypasses the meter coil. Such an instrument is called an ammeter. If the coil and any additional series resistance included obey Ohm's law, the meter can also be calibrated to read potential difference or voltage. The instrument is then called a voltmeter. A good ammeter has very low resistance; a good voltmeter has very high resistance. (See Examples 26.8–26.11.)



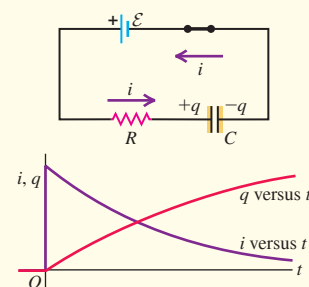
R-C circuits: When a capacitor is charged by a battery in series with a resistor, the current and capacitor charge are not constant. The charge approaches its final value asymptotically and the current approaches zero asymptotically. The charge and current in the circuit are given by Eqs. (26.12) and (26.13). After a time $\tau = RC$, the charge has approached within $1/e$ of its final value. This time is called the time constant or relaxation time of the circuit. When the capacitor discharges, the charge and current are given as functions of time by Eqs. (26.16) and (26.17). The time constant is the same for charging and discharging. (See Examples 26.12 and 26.13.)

Capacitor charging:

$$q = C\mathcal{E}(1 - e^{-t/RC}) \quad (26.12)$$

$$= Q_f(1 - e^{-t/RC})$$

$$i = \frac{dq}{dt} = \frac{\mathcal{E}}{R}e^{-t/RC} = I_0e^{-t/RC} \quad (26.13)$$

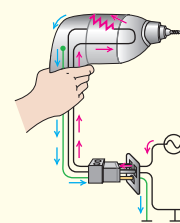


Capacitor discharging:

$$q = Q_0e^{-t/RC} \quad (26.16)$$

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC}e^{-t/RC} = I_0e^{-t/RC} \quad (26.17)$$

Household wiring: In household wiring systems, the various electrical devices are connected in parallel across the power line, which consists of a pair of conductors, one “hot” and the other “neutral.” An additional “ground” wire is included for safety. The maximum permissible current in a circuit is determined by the size of the wires and the maximum temperature they can tolerate. Protection against excessive current and the resulting fire hazard is provided by fuses or circuit breakers. (See Example 26.14.)





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 26.1 and 26.2** (Section 26.1) before attempting these problems.

VP26.2.1 You have three resistors: $R_1 = 1.00\ \Omega$, $R_2 = 2.00\ \Omega$, and $R_3 = 4.00\ \Omega$. Find the equivalent resistance for the combinations shown in (a) Fig. 26.1a, (b) Fig. 26.1b, (c) Fig. 26.1c, and (d) Fig. 26.1d.

VP26.2.2 You have three resistors, $R_1 = 4.00\ \Omega$, $R_2 = 5.00\ \Omega$, and $R_3 = 6.00\ \Omega$, connected as shown in Fig. 26.1d. The terminals of a battery with emf 24.0 V and negligible internal resistance are connected to points a and b . Find the current through (a) the battery, (b) R_1 , (c) R_2 , and (d) R_3 .

VP26.2.3 You have three resistors, $R_1 = 7.00\ \Omega$, $R_2 = 8.00\ \Omega$, and $R_3 = 9.00\ \Omega$, connected as shown in Fig. 26.1b. The terminals of a battery with emf 12.0 V and negligible internal resistance are connected to points a and b . Find (a) the power output of the battery and the power input to (b) R_1 , (c) R_2 , and (d) R_3 .

VP26.2.4 You have three resistors, $R_1 = 5.00\ \Omega$, $R_2 = 6.00\ \Omega$, and $R_3 = 7.00\ \Omega$, connected as shown in Fig. 26.1c. The terminals of a battery with emf 9.00 V and negligible internal resistance are connected to points a and b . Find (a) the power output of the battery and the power input to (b) R_1 , (c) R_2 , and (d) R_3 .

Be sure to review **EXAMPLES 26.3, 26.4, 26.5, 26.6, and 26.7** (Section 26.2) before attempting these problems.

VP26.7.1 In the circuit shown in Fig. 26.10a, you reverse the 4 V battery so that its positive terminal is on the right side instead of the left. Find (a) the current in the circuit, (b) the potential difference V_{ab} , and (c) the power output of the emf of each battery.

VP26.7.2 Figure 26.6a shows two batteries, one with emf $\mathcal{E}_1$ and one with emf $\mathcal{E}_2$, connected to a resistance R . Current I_1 flows through emf $\mathcal{E}_1$ from point c toward point a , and current I_2 flows through emf $\mathcal{E}_2$ from point b toward point a . If $\mathcal{E}_1 = 8.00\ \text{V}$, $\mathcal{E}_2 = 9.00\ \text{V}$, $R = 5.00\ \Omega$, $I_1 = 0.200\ \text{A}$, and $I_2 = 1.35\ \text{A}$, find the values of (a) the potential difference V_{ab} , (b) the internal resistance r_1 , and (c) the internal resistance r_2 .

VP26.7.3 In the circuit shown in Figure 26.12, you replace the $2\ \Omega$ resistor between points b and d with a $3\ \Omega$ resistor. Find the power dissipated in (a) the $1\ \Omega$ resistor between points c and a , (b) the $1\ \Omega$ resistor between points c and b , (c) the $1\ \Omega$ resistor between points a and b , (d) the $1\ \Omega$ resistor between points a and d , and (e) the $3\ \Omega$ resistor between points b and d . Give each answer to three significant figures.

VP26.7.4 In the circuit shown in Fig. 26.9a, let $r_1 = r_2 = 0$ (that is, the internal resistances are very small) and let $R_1 = R_2 = R_3 = R$. Find the values of (a) I_1 , (b) I_2 , and (c) I_3 .

Be sure to review **EXAMPLES 26.12 and 26.13** (Section 26.4) before attempting these problems.

VP26.13.1 You connect a $10.0\ \text{M}\Omega$ resistor in series with a $3.20\ \mu\text{F}$ capacitor and a battery with emf 9.00 V. Before you close the switch at $t = 0$ to complete the circuit, the capacitor is uncharged. Find (a) the final capacitor charge, (b) the initial current, (c) the time constant, (d) the fraction of the final charge on the capacitor at $t = 18.0\ \text{s}$, and (e) the fraction of the initial current present at $t = 18.0\ \text{s}$.

VP26.13.2 A $2.20\ \mu\text{F}$ capacitor initially has charge $4.20\ \mu\text{C}$. You connect it in series with a $4.00\ \text{M}\Omega$ resistor and an open switch. If $t = 0$ is the time when you close the switch, (a) at what time will the capacitor charge be $1.20\ \mu\text{C}$ and (b) what will be the current in the circuit at this time?

VP26.13.3 You connect an unknown resistor in series with an $8.00\ \mu\text{F}$ capacitor and an open switch. Before you close the switch at $t = 0$, the capacitor has charge $5.50\ \mu\text{C}$; at $t = 17.0\ \text{s}$, the charge has decreased to $1.10\ \mu\text{C}$. Find (a) the resistance of the resistor, (b) the current in the circuit just after you close the switch, and (c) the current in the circuit at $t = 17.0\ \text{s}$.

VP26.13.4 You connect an initially uncharged $6.40\ \mu\text{F}$ capacitor in series with a $5.00\ \text{M}\Omega$ resistor and a battery with emf 12.0 V. After letting the capacitor charge for 51.0 s, you disconnect it from this circuit and connect it in series to an open switch and a $6.00\ \text{M}\Omega$ resistor. Find the charge on the capacitor (a) when you disconnect it from the first circuit and (b) 70.0 s after you close the switch in the second circuit.

BRIDGING PROBLEM Two Capacitors and Two Resistors

A $2.40\ \mu\text{F}$ capacitor and a $3.60\ \mu\text{F}$ capacitor are connected in series. (a) A charge of $5.20\ \text{mC}$ is placed on each capacitor. What is the energy stored in the capacitors? (b) A $655\ \Omega$ resistor is connected to the terminals of the capacitor combination, and a voltmeter with resistance $4.58 \times 10^4\ \Omega$ is connected across the resistor (Fig. 26.27). What is the rate of change of the energy stored in the capacitors just after the connection is made? (c) How long after the connection is made has the energy stored in the capacitors decreased to $1/e$ of its initial value? (d) At the instant calculated in part (c), what is the rate of change of the energy stored in the capacitors?

SOLUTION GUIDE

IDENTIFY and SET UP

1. The two capacitors act as a single equivalent capacitor (see Section 24.2), and the resistor and voltmeter act as a single equivalent resistor. Select equations that will allow you to calculate the values of these equivalent circuit elements.
2. In part (a) you'll need to use Eq. (24.9), which gives the energy stored in a capacitor.
3. For parts (b), (c), and (d), you'll need to use Eq. (24.9) as well as Eqs. (26.16) and (26.17), which give the capacitor charge and

current as functions of time. (Hint: The rate at which energy is lost by the capacitors equals the rate at which energy is dissipated in the resistances.)

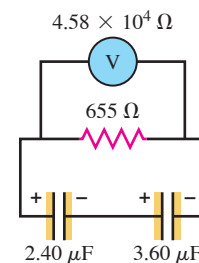
Figure 26.27 When the connection is made, the charged capacitors discharge.

EXECUTE

4. Find the stored energy at $t = 0$.
5. Find the rate of change of the stored energy at $t = 0$.
6. Find the value of t at which the stored energy has $1/e$ of the value you found in step 4.
7. Find the rate of change of the stored energy at the time you found in step 6.

EVALUATE

8. Check your results from steps 5 and 7 by calculating the rate of change in a different way. (Hint: The rate of change of the stored energy U is dU/dt .)



PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

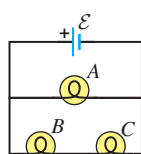
Q26.1 In which 120 V light bulb does the filament have greater resistance: a 60 W bulb or a 120 W bulb? If the two bulbs are connected to a 120 V line in series, through which bulb will there be the greater voltage drop? What if they are connected in parallel? Explain your reasoning.

Q26.2 Two 120 V light bulbs, one 25 W and one 200 W, were connected in series across a 240 V line. It seemed like a good idea at the time, but one bulb burned out almost immediately. Which one burned out, and why?

Q26.3 You connect a number of identical light bulbs to a flashlight battery. (a) What happens to the brightness of each bulb as more and more bulbs are added to the circuit if you connect them (i) in series and (ii) in parallel? (b) Will the battery last longer if the bulbs are in series or in parallel? Explain your reasoning.

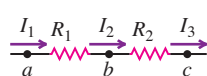
Q26.4 In the circuit shown in **Fig. Q26.4**, three identical light bulbs are connected to a flashlight battery. How do the brightnesses of the bulbs compare? Which light bulb has the greatest current passing through it? Which light bulb has the greatest potential difference between its terminals? What happens if bulb A is unscrewed? Bulb B? Bulb C? Explain your reasoning.

Figure Q26.4



Q26.5 If two resistors R_1 and R_2 ($R_2 > R_1$) are connected in series as shown in **Fig. Q26.5**, which of the following must be true? In each case justify your answer. (a) $I_1 = I_2 = I_3$. (b) The current is greater in R_1 than in R_2 .

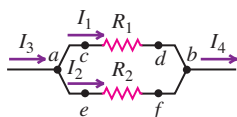
Figure Q26.5



(c) The electrical power consumption is the same for both resistors. (d) The electrical power consumption is greater in R_2 than in R_1 . (e) The potential drop is the same across both resistors. (f) The potential at point a is the same as at point c. (g) The potential at point b is lower than at point c. (h) The potential at point c is lower than at point b.

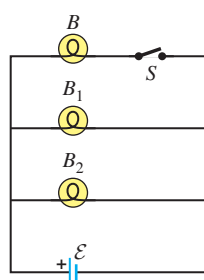
Q26.6 If two resistors R_1 and R_2 ($R_2 > R_1$) are connected in parallel as shown in **Fig. Q26.6**, which of the following must be true? In each case justify your answer. (a) $I_1 = I_2$. (b) $I_3 = I_4$. (c) The current is greater in R_1 than in R_2 . (d) The rate of electrical energy consumption is the same for both resistors. (e) The rate of electrical energy consumption is greater in R_2 than in R_1 . (f) $V_{cd} = V_{ef} = V_{ab}$. (g) Point c is at higher potential than point d. (h) Point f is at higher potential than point e. (i) Point c is at higher potential than point e.

Figure Q26.6



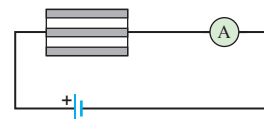
Q26.7 A battery with no internal resistance is connected across identical light bulbs as shown in **Fig. Q26.7**. When you close the switch S , will the brightness of bulbs B_1 and B_2 change? If so, how will it change? Explain.

Figure Q26.7



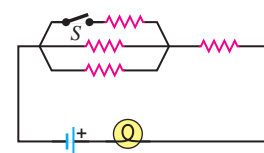
Q26.8 A resistor consists of three identical metal strips connected as shown in **Fig. Q26.8**. If one of the strips is cut out, does the ammeter reading increase, decrease, or stay the same? Why?

Figure Q26.8



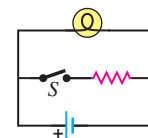
Q26.9 A light bulb is connected in the circuit shown in **Fig. Q26.9**. If we close the switch S , does the bulb's brightness increase, decrease, or remain the same? Explain why.

Figure Q26.9



Q26.10 A real battery, having nonnegligible internal resistance, is connected across a light bulb as shown in **Fig. Q26.10**. When the switch S is closed, what happens to the brightness of the bulb? Why?

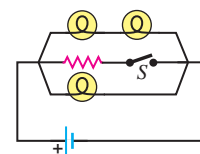
Figure Q26.10



Q26.11 If the battery in Discussion Question Q26.10 is ideal with no internal resistance, what will happen to the brightness of the bulb when S is closed? Why?

Q26.12 Consider the circuit shown in **Fig. Q26.12**. What happens to the brightnesses of the bulbs when the switch S is closed if the battery (a) has no internal resistance and (b) has nonnegligible internal resistance? Explain why.

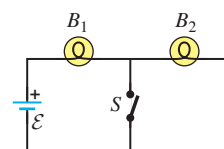
Figure Q26.12



Q26.13 Is it possible to connect resistors together in a way that cannot be reduced to some combination of series and parallel combinations? If so, give examples. If not, state why not.

Q26.14 The battery in the circuit shown in **Fig. Q26.14** has no internal resistance. After you close the switch S , will the brightness of bulb B_1 increase, decrease, or stay the same?

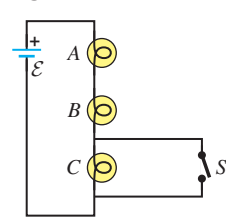
Figure Q26.14



Q26.15 In a two-cell flashlight, the batteries are usually connected in series. Why not connect them in parallel? What possible advantage could there be in connecting several identical batteries in parallel?

Q26.16 Identical light bulbs A , B , and C are connected as shown in **Fig. Q26.16**. When the switch S is closed, bulb C goes out. Explain why. What happens to the brightness of bulbs A and B ? Explain.

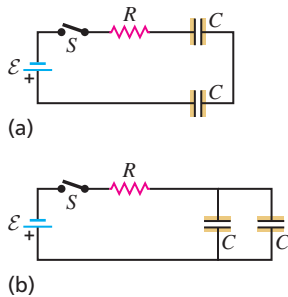
Figure Q26.16



Q26.17 The emf of a flashlight battery is roughly constant with time, but its internal resistance increases with age and use. What sort of meter should be used to test the freshness of a battery?

Q26.18 Will the capacitors in the circuits shown in **Fig. Q26.18** charge at the same rate when the switch S is closed? If not, in which circuit will the capacitors charge more rapidly? Explain.

Figure Q26.18



Q26.19 Verify that the time constant RC has units of time.

Q26.20 For very large resistances it is easy to construct R - C circuits that have time constants of several seconds or minutes. How might this fact be used to measure very large resistances, those that are too large to measure by more conventional means?

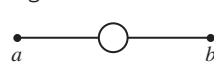
Q26.21 When a capacitor, battery, and resistor are connected in series, does the resistor affect the maximum charge stored on the capacitor? Why or why not? What purpose does the resistor serve?

EXERCISES

Section 26.1 Resistors in Series and Parallel

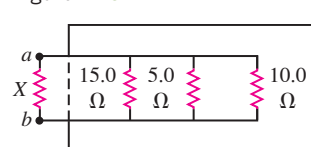
26.1 • A uniform wire of resistance R is cut into three equal lengths. One of these is formed into a circle and connected between the other two (**Fig. E26.1**). What is the resistance between the opposite ends a and b ?

Figure E26.1



26.2 • A machine part has a resistor X protruding from an opening in the side. This resistor is connected to three other resistors, as shown in **Fig. E26.2**. An ohmmeter connected across a and b reads $2.00\ \Omega$. What is the resistance of X ?

Figure E26.2

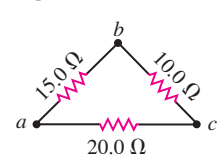


26.3 • A resistor with $R_1 = 25.0\ \Omega$ is connected to a battery that has negligible internal resistance and electrical energy is dissipated by R_1 at a rate of $36.0\ \text{W}$. If a second resistor with $R_2 = 15.0\ \Omega$ is connected in series with R_1 , what is the total rate at which electrical energy is dissipated by the two resistors?

26.4 • A $42\ \Omega$ resistor and a $20\ \Omega$ resistor are connected in parallel, and the combination is connected across a $240\ \text{V}$ dc line. (a) What is the resistance of the parallel combination? (b) What is the total current through the parallel combination? (c) What is the current through each resistor?

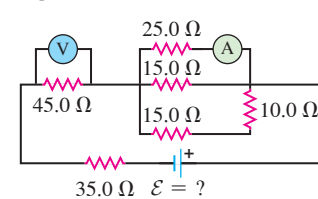
26.5 • A triangular array of resistors is shown in **Fig. E26.5**. What current will this array draw from a $35.0\ \text{V}$ battery having negligible internal resistance if we connect it across (a) ab ; (b) bc ; (c) ac ? (d) If the battery has an internal resistance of $3.00\ \Omega$, what current will the array draw if the battery is connected across bc ?

Figure E26.5



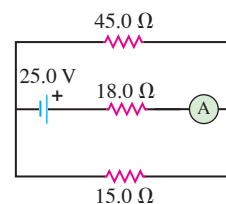
26.6 • For the circuit shown in **Fig. E26.6** both meters are idealized, the battery has no appreciable internal resistance, and the ammeter reads $1.25\ \text{A}$. (a) What does the voltmeter read? (b) What is the emf $\mathcal{E}$ of the battery?

Figure E26.6



26.7 • For the circuit shown in **Fig. E26.7** find the reading of the idealized ammeter if the battery has an internal resistance of $3.26\ \Omega$.

Figure E26.7



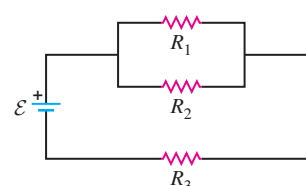
26.8 • Two resistors, R_1 and R_2 , are connected in parallel to a power supply that has voltage V and negligible internal resistance. $R_2 = 8.00\ \Omega$ and the resistance of R_1 is not known. For several values of V , you measure the current I flowing through the voltage source. You plot the data as I versus V and find that they lie close to a straight line that has slope $0.208\ \Omega^{-1}$. What is the resistance of R_1 ?

26.9 • Six identical resistors, each with resistance R , are connected to an emf $\mathcal{E}$. (a) In terms of $\mathcal{E}$ and R , what is the current I through each of the resistors if they are connected in parallel? (b) In series? (c) For which network of resistors, series or parallel, is the power consumed in each resistor greater?

26.10 • **Power Rating of a Resistor.** The *power rating* of a resistor is the maximum power the resistor can safely dissipate without too great a rise in temperature and hence damage to the resistor. (a) If the power rating of a $15\ \text{k}\Omega$ resistor is $5.0\ \text{W}$, what is the maximum allowable potential difference across the terminals of the resistor? (b) A $9.0\ \text{k}\Omega$ resistor is to be connected across a $120\ \text{V}$ potential difference. What power rating is required? (c) A $100.0\ \Omega$ and a $150.0\ \Omega$ resistor, both rated at $2.00\ \text{W}$, are connected in series across a variable potential difference. What is the greatest this potential difference can be without overheating either resistor, and what is the rate of heat generated in each resistor under these conditions?

26.11 • In **Fig. E26.11**, $R_1 = 3.00\ \Omega$, $R_2 = 6.00\ \Omega$, and $R_3 = 5.00\ \Omega$. The battery has negligible internal resistance. The current I_2 through R_2 is $4.00\ \text{A}$. (a) What are the currents I_1 and I_3 ? (b) What is the emf of the battery?

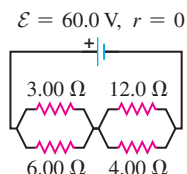
Figure E26.11



26.12 • In **Fig. E26.11** the battery has emf $35.0\ \text{V}$ and negligible internal resistance. $R_1 = 5.00\ \Omega$. The current through R_1 is $1.50\ \text{A}$, and the current through $R_3 = 4.50\ \text{A}$. What are the resistances R_2 and R_3 ?

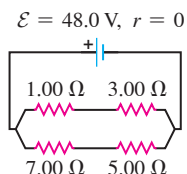
26.13 • Compute the equivalent resistance of the network in **Fig. E26.13**, and find the current in each resistor. The battery has negligible internal resistance.

Figure E26.13



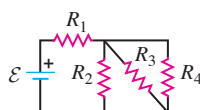
26.14 • Compute the equivalent resistance of the network in **Fig. E26.14**, and find the current in each resistor. The battery has negligible internal resistance.

Figure E26.14



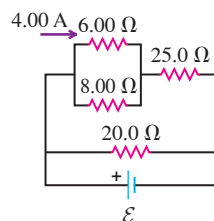
26.15 • In the circuit of **Fig. E26.15**, each resistor represents a light bulb. Let $R_1 = R_2 = R_3 = R_4 = 4.50 \, \Omega$ and $\mathcal{E} = 9.00 \, \text{V}$. (a) Find the current in each bulb. (b) Find the power dissipated in each bulb. Which bulb or bulbs glow the brightest? (c) Bulb R_4 is now removed from the circuit, leaving a break in the wire at its position. Now what is the current in each of the remaining bulbs R_1 , R_2 , and R_3 ? (d) With bulb R_4 removed, what is the power dissipated in each of the remaining bulbs? (e) Which light bulb(s) glow brighter as a result of removing R_4 ? Which bulb(s) glow less brightly? Discuss why there are different effects on different bulbs.

Figure E26.15



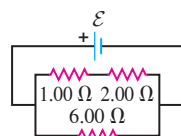
26.16 • Consider the circuit shown in **Fig. E26.16**. The current through the $6.00 \, \Omega$ resistor is $4.00 \, \text{A}$, in the direction shown. What are the currents through the $25.0 \, \Omega$ and $20.0 \, \Omega$ resistors?

Figure E26.16



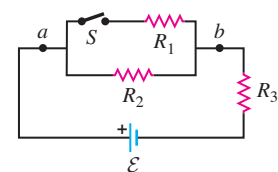
26.17 • In the circuit shown in **Fig. E26.17**, the voltage across the $2.00 \, \Omega$ resistor is $12.0 \, \text{V}$. What are the emf of the battery and the current through the $6.00 \, \Omega$ resistor?

Figure E26.17



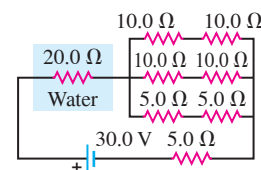
26.18 •• In the circuit shown in **Fig. E26.18**, $\mathcal{E} = 36.0 \, \text{V}$, $R_1 = 4.00 \, \Omega$, $R_2 = 6.00 \, \Omega$, and $R_3 = 3.00 \, \Omega$. (a) What is the potential difference V_{ab} between points a and b when the switch S is open and when S is closed? (b) For each resistor, calculate the current through the resistor with S open and with S closed. For each resistor, does the current increase or decrease when S is closed?

Figure E26.18



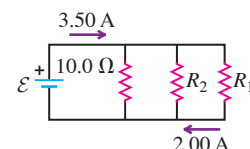
26.19 •• CP In the circuit in **Fig. E26.19**, a $20.0 \, \Omega$ resistor is inside $100 \, \text{g}$ of pure water that is surrounded by insulating Styrofoam. If the water is initially at 10.0°C , how long will it take for its temperature to rise to 58.0°C ?

Figure E26.19



26.20 • In the circuit shown in **Fig. E26.20**, the rate at which R_1 is dissipating electrical energy is $15.0 \, \text{W}$. (a) Find R_1 and R_2 . (b) What is the emf of the battery? (c) Find the current through both R_2 and the $10.0 \, \Omega$ resistor. (d) Calculate the total electrical power consumption in all the resistors and the electrical power delivered by the battery. Show that your results are consistent with conservation of energy.

Figure E26.20

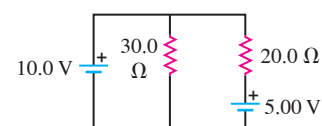


26.21 • Light Bulbs in Series and in Parallel. Two light bulbs have constant resistances of $400 \, \Omega$ and $800 \, \Omega$. If the two light bulbs are connected in series across a $120 \, \text{V}$ line, find (a) the current through each bulb; (b) the power dissipated in each bulb; (c) the total power dissipated in both bulbs. The two light bulbs are now connected in parallel across the $120 \, \text{V}$ line. Find (d) the current through each bulb; (e) the power dissipated in each bulb; (f) the total power dissipated in both bulbs. (g) In each situation, which of the two bulbs glows the brightest? (h) In which situation is there a greater total light output from both bulbs combined?

Section 26.2 Kirchhoff's Rules

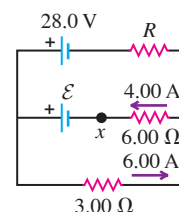
26.22 •• The batteries shown in the circuit in **Fig. E26.22** have negligibly small internal resistances. Find the current through (a) the $30.0 \, \Omega$ resistor; (b) the $20.0 \, \Omega$ resistor; (c) the $10.0 \, \text{V}$ battery.

Figure E26.22



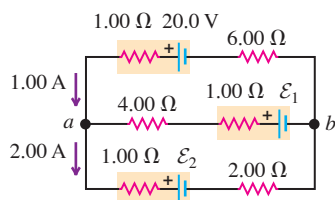
26.23 • In the circuit shown in **Fig. E26.23** find (a) the current in resistor R ; (b) the resistance R ; (c) the unknown emf $\mathcal{E}$. (d) If the circuit is broken at point x , what is the current in resistor R ?

Figure E26.23



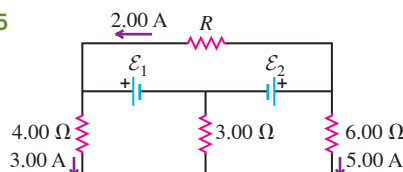
26.24 • Find the emfs $\mathcal{E}_1$ and $\mathcal{E}_2$ in the circuit of **Fig. E26.24**, and find the potential difference of point b relative to point a .

Figure E26.24



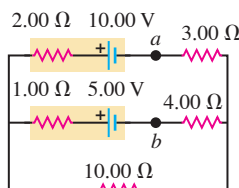
26.25 • In the circuit shown in **Fig. E26.25**, find (a) the current in the $3.00\ \Omega$ resistor; (b) the unknown emfs $\mathcal{E}_1$ and $\mathcal{E}_2$; (c) the resistance R . Note that three currents are given.

Figure E26.25



26.26 • In the circuit shown in **Fig. E26.26**, find (a) the current in each branch and (b) the potential difference V_{ab} of point a relative to point b .

Figure E26.26

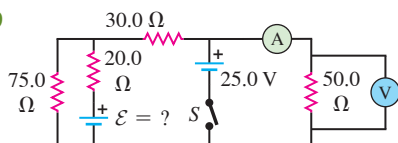


26.27 • The $10.00\ \text{V}$ battery in **Fig. E26.26** is removed from the circuit and reinserted with the opposite polarity, so that its positive terminal is now next to point a . The rest of the circuit is as shown in the figure. Find (a) the current in each branch and (b) the potential difference V_{ab} of point a relative to point b .

26.28 • The $5.00\ \text{V}$ battery in **Fig. E26.26** is removed from the circuit and replaced by a $15.00\ \text{V}$ battery, with its negative terminal next to point b . The rest of the circuit is as shown in the figure. Find (a) the current in each branch and (b) the potential difference V_{ab} of point a relative to point b .

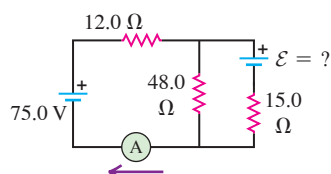
26.29 • In the circuit shown in **Fig. E26.29** the batteries have negligible internal resistance and the meters are both idealized. With the switch S open, the voltmeter reads $15.0\ \text{V}$. (a) Find the emf $\mathcal{E}$ of the battery. (b) What will the ammeter read when the switch is closed?

Figure E26.29



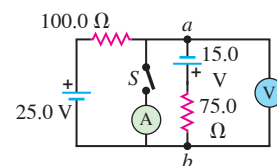
26.30 • In the circuit shown in **Fig. E26.30** both batteries have insignificant internal resistance and the idealized ammeter reads $1.50\ \text{A}$ in the direction shown. Find the emf $\mathcal{E}$ of the battery. Is the polarity shown correct?

Figure E26.30



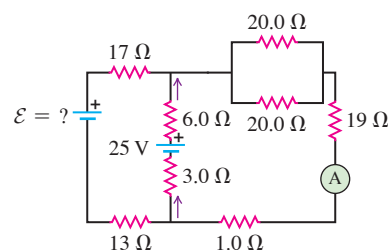
26.31 • In the circuit shown in **Fig. E26.31** all meters are idealized and the batteries have no appreciable internal resistance. (a) Find the reading of the voltmeter with the switch S open. Which point is at a higher potential: a or b ? (b) With S closed, find the reading of the voltmeter and the ammeter. Which way (up or down) does the current flow through the switch?

Figure E26.31



26.32 • In the circuit shown in **Fig. E26.32**, the $6.0\ \Omega$ resistor is consuming energy at a rate of $24\ \text{J/s}$ when the current through it flows as shown. (a) Find the current through the ammeter A . (b) What are the polarity and emf $\mathcal{E}$ of the unknown battery, assuming it has negligible internal resistance?

Figure E26.32

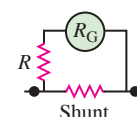


Section 26.3 Electrical Measuring Instruments

26.33 • The resistance of a galvanometer coil is $25.0\ \Omega$, and the current required for full-scale deflection is $500\ \mu\text{A}$. (a) Show in a diagram how to convert the galvanometer to an ammeter reading $20.0\ \text{mA}$ full scale, and compute the shunt resistance. (b) Show how to convert the galvanometer to a voltmeter reading $500\ \text{mV}$ full scale, and compute the series resistance.

26.34 • The resistance of the coil of a pivoted-coil galvanometer is $9.36\ \Omega$, and a current of $0.0224\ \text{A}$ causes it to deflect full scale. We want to convert this galvanometer to an ammeter reading $20.0\ \text{A}$ full scale. The only shunt available has a resistance of $0.0250\ \Omega$. What resistance R must be connected in series with the coil (**Fig. E26.34**)?

Figure E26.34



26.35 • A circuit consists of a series combination of $6.00\ \text{k}\Omega$ and $5.00\ \text{k}\Omega$ resistors connected across a $50.0\ \text{V}$ battery having negligible internal resistance. You want to measure the true potential difference (that is, the potential difference without the meter present) across the $5.00\ \text{k}\Omega$ resistor using a voltmeter having an internal resistance of $10.0\ \text{k}\Omega$. (a) What potential difference does the voltmeter measure across the $5.00\ \text{k}\Omega$ resistor? (b) What is the true potential difference across this resistor when the meter is not present? (c) By what percentage is the voltmeter reading in error from the true potential difference?

Section 26.4 R-C Circuits

26.36 • To measure the capacitance C of a capacitor, you attach the capacitor to a battery and wait until it is fully charged. You then disconnect the capacitor from the battery and let it discharge through a resistor of resistance R . You measure the time $T_{1/2}$ that it takes the voltage across the resistor to decrease to half its initial value at the instant that the connection to the capacitor is first completed. You repeat this for several different resistors. You plot the data as $T_{1/2}$ versus R and find that they lie close to a straight line that has slope $5.00\ \mu\text{F}$. What is the capacitance C of the capacitor?

26.37 • A capacitor is charged to a potential of $12.0\ \text{V}$ and is then connected to a voltmeter having a total internal resistance of $3.40\ \text{M}\Omega$. After a time of $4.00\ \text{s}$ the voltmeter reads $3.0\ \text{V}$. What are (a) the capacitance and (b) the time constant of the circuit?

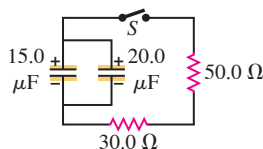
26.38 •• You connect a battery, resistor, and capacitor as in Fig. 26.20a, where $\mathcal{E} = 36.0 \text{ V}$, $C = 5.00 \mu\text{F}$, and $R = 120 \Omega$. The switch S is closed at $t = 0$. (a) When the voltage across the capacitor is 8.00 V , what is the magnitude of the current in the circuit? (b) At what time t after the switch is closed is the voltage across the capacitor 8.00 V ? (c) When the voltage across the capacitor is 8.00 V , at what rate is energy being stored in the capacitor?

26.39 • A $4.60 \mu\text{F}$ capacitor that is initially uncharged is connected in series with a $7.50 \text{ k}\Omega$ resistor and an emf source with $\mathcal{E} = 245 \text{ V}$ and negligible internal resistance. Just after the circuit is completed, what are (a) the voltage drop across the capacitor; (b) the voltage drop across the resistor; (c) the charge on the capacitor; (d) the current through the resistor? (e) A long time after the circuit is completed (after many time constants) what are the values of the quantities in parts (a)–(d)?

26.40 •• You connect a battery, resistor, and capacitor as in Fig. 26.20a, where $R = 12.0 \Omega$ and $C = 5.00 \times 10^{-6} \text{ F}$. The switch S is closed at $t = 0$. When the current in the circuit has magnitude 3.00 A , the charge on the capacitor is $40.0 \times 10^{-6} \text{ C}$. (a) What is the emf of the battery? (b) At what time t after the switch is closed is the charge on the capacitor equal to $40.0 \times 10^{-6} \text{ C}$? (c) When the current has magnitude 3.00 A , at what rate is energy being (i) stored in the capacitor, (ii) supplied by the battery?

26.41 •• CP In the circuit shown in Fig. E26.41 both capacitors are initially charged to 45.0 V . (a) How long after closing the switch S will the potential across each capacitor be reduced to 10.0 V , and (b) what will be the current at that time?

Figure E26.41



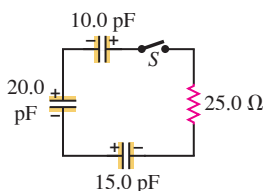
26.42 • A $12.4 \mu\text{F}$ capacitor is connected through a $0.895 \text{ M}\Omega$ resistor to a constant potential difference of 60.0 V . (a) Compute the charge on the capacitor at the following times after the connections are made: 0 , 5.0 s , 10.0 s , 20.0 s , and 100.0 s . (b) Compute the charging currents at the same instants. (c) Graph the results of parts (a) and (b) for t between 0 and 20 s .

26.43 • An emf source with $\mathcal{E} = 120 \text{ V}$, a resistor with $R = 80.0 \Omega$, and a capacitor with $C = 4.00 \mu\text{F}$ are connected in series. As the capacitor charges, when the current in the resistor is 0.900 A , what is the magnitude of the charge on each plate of the capacitor?

26.44 • A network of two identical capacitors, each with capacitance C , is charged through a resistor R using a battery with emf $\mathcal{E}$. (a) What is the time constant, in terms of R and C , for the charging circuit if the two capacitors are in series? (b) In parallel? (c) For which capacitor network, series or parallel, does the voltage across the resistor become 1% of its initial value in a shorter time?

26.45 •• CP In the circuit shown in Fig. E26.45 each capacitor initially has a charge of magnitude 3.50 nC on its plates. After the switch S is closed, what will be the current in the circuit at the instant that the capacitors have lost 80.0% of their initial stored energy?

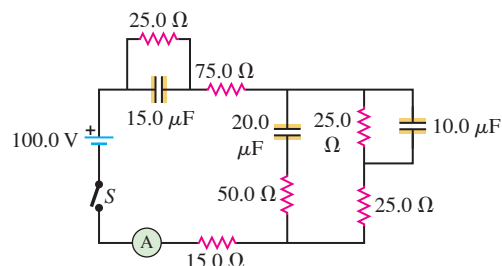
Figure E26.45



26.46 • A $1.50 \mu\text{F}$ capacitor is charging through a 12.0Ω resistor using a 10.0 V battery. What will be the current when the capacitor has acquired $\frac{1}{4}$ of its maximum charge? Will it be $\frac{1}{4}$ of the maximum current?

26.47 • In the circuit in Fig. E26.47 the capacitors are initially uncharged, the battery has no internal resistance, and the ammeter is idealized. Find the ammeter reading (a) just after the switch S is closed and (b) after S has been closed for a very long time.

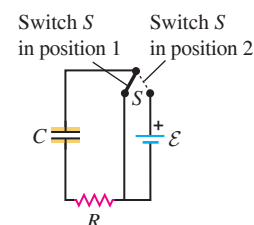
Figure E26.47



26.48 • A $12.0 \mu\text{F}$ capacitor is charged to a potential of 50.0 V and then discharged through a 225Ω resistor. How long does it take the capacitor to lose (a) half of its charge and (b) half of its stored energy?

26.49 • In the circuit shown in Fig. E26.49, $C = 5.90 \mu\text{F}$, $\mathcal{E} = 28.0 \text{ V}$, and the emf has negligible resistance. Initially the capacitor is uncharged and the switch S is in position 1. The switch is then moved to position 2, so that the capacitor begins to charge. (a) What will be the charge on the capacitor a long time after S is moved to position 2? (b) After S has been in position 2 for 3.00 ms , the charge on the capacitor is measured to be $110 \mu\text{C}$. What is the value of the resistance R ? (c) How long after S is moved to position 2 will the charge on the capacitor be equal to 99.0% of the final value found in part (a)?

Figure E26.49



26.50 • A capacitor with $C = 6.00 \mu\text{F}$ is fully charged by connecting it to a battery that has emf 50.0 V . The capacitor is disconnected from the battery. A resistor of resistance $R = 185 \Omega$ is connected across the capacitor, and the capacitor discharges through the resistor. (a) What is the charge q on the capacitor when the current in the resistor is 0.180 A ? (b) If the connection to the resistor is completed at time $t = 0$, what is the value of t when the current has the value specified in part (a)?

Section 26.5 Power Distribution Systems

26.51 • A 1500 W electric heater is plugged into the outlet of a 120 V circuit that has a 20 A circuit breaker. You plug an electric hair dryer into the same outlet. The hair dryer has power settings of 600 W , 900 W , 1200 W , and 1500 W . You start with the hair dryer on the 600 W setting and increase the power setting until the circuit breaker trips. What power setting caused the breaker to trip?

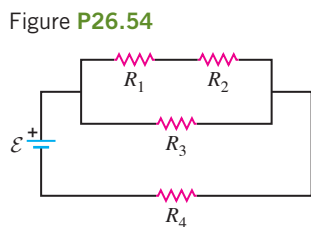
26.52 • The heating element of an electric dryer is rated at 4.1 kW when connected to a 240 V line. (a) What is the current in the heating element? Is 12 gauge wire large enough to supply this current? (b) What is the resistance of the dryer's heating element at its operating temperature? (c) At 11 cents per kWh , how much does it cost per hour to operate the dryer?

PROBLEMS

26.53 •• CALC A capacitor with capacitance C is connected in series to a resistor of resistance R and a battery with emf $\mathcal{E}$. The circuit is completed at time $t = 0$. (a) In terms of $\mathcal{E}$, R , and C , how much energy is stored in the capacitor when it is fully charged? (b) The power output of the battery is $P_{\mathcal{E}} = \mathcal{E}i$, with i given by Eq. (26.13). The electrical energy supplied in an infinitesimal time interval dt is $P_{\mathcal{E}}dt$. Integrate from $t = 0$ to $t \rightarrow \infty$ to find the total energy supplied by the battery. (c) The rate of consumption of electrical energy in the resistor is $P_R = i^2R$. In an infinitesimal time interval dt , the amount of electrical energy consumed by the resistor is P_Rdt . Integrate from $t = 0$ to $t \rightarrow \infty$ to find the total energy consumed by the resistor. (d) What fraction of the total energy supplied by the battery is stored in the capacitor? What fraction is consumed in the resistor?

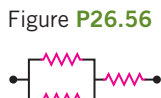
26.54 •• In **Fig. P26.54**, the battery has negligible internal resistance and $\mathcal{E} = 48.0$ V.

$R_1 = R_2 = 4.00 \Omega$ and $R_4 = 3.00 \Omega$. What must the resistance R_3 be for the resistor network to dissipate electrical energy at a rate of 295 W?



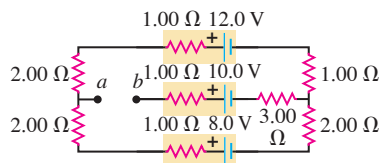
26.55 •• Consider the circuit shown in **Fig. 26.6a**. $\mathcal{E}_1 = 24.0$ V, $\mathcal{E}_2 = 36.0$ V, $r_1 = r_2 = 2.00 \Omega$, and $R = 20.0 \Omega$. (a) What is the current I_1 through $\mathcal{E}_1$? (b) What is the current I_2 through $\mathcal{E}_2$? (c) What is the current I_R through R ?

26.56 •• Each of the three resistors in **Fig. P26.56** has a resistance of 2.4Ω and can dissipate a maximum of 48 W without becoming excessively heated. What is the maximum power the circuit can dissipate?



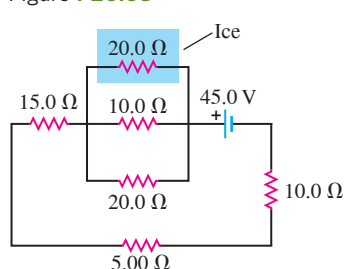
26.57 •• (a) Find the potential of point a with respect to point b in **Fig. P26.57**. (b) If points a and b are connected by a wire with negligible resistance, find the current in the 12.0 V battery.

Figure P26.57



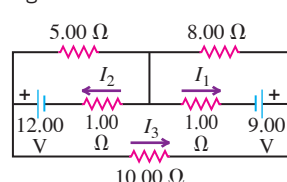
26.58 •• CP For the circuit shown in **Fig. P26.58** a 20.0Ω resistor is embedded in a large block of ice at 0.00°C , and the battery has negligible internal resistance. At what rate (in g/s) is this circuit melting the ice? (The latent heat of fusion for ice is 3.34×10^5 J/kg.)

Figure P26.58



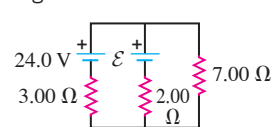
26.59 • Calculate the three currents I_1 , I_2 , and I_3 indicated in the circuit diagram shown in **Fig. P26.59**.

Figure P26.59



26.60 ••• What must the emf $\mathcal{E}$ in **Fig. P26.60** be in order for the current through the 7.00Ω resistor to be 1.80 A? Each emf source has negligible internal resistance.

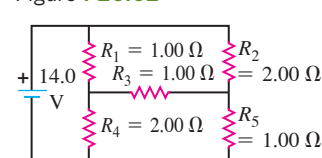
Figure P26.60



26.61 •• A standard phone charger provides 5.0 V of potential and supplies 1.0 A of current to the phone. (a) Estimate the diameter of each of the two wires inside your charge cable. (b) Assume the wire is copper, which has a resistivity of $1.72 \times 10^{-8} \Omega \cdot \text{m}$, and determine its resistance per meter. (c) If the phone draws $I = 1.0$ A, the potential supplied to the phone will be lower than the potential on the side of the charger by an amount IR , where R is the resistance of the wire. Determine the length of charging wire that would diminish the potential to 4.0 V at the phone.

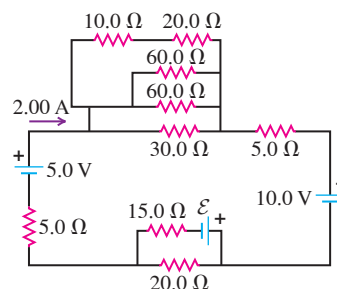
26.62 • (a) Find the current through the battery and each resistor in the circuit shown in **Fig. P26.62**. (b) What is the equivalent resistance of the resistor network?

Figure P26.62



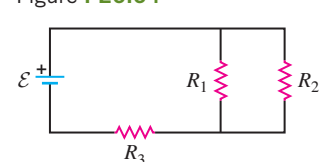
26.63 •• Consider the circuit shown in **Fig. P26.63**. (a) What must the emf $\mathcal{E}$ of the battery be in order for a current of 2.00 A to flow through the 5.00 V battery as shown? Is the polarity of the battery correct as shown? (b) How long does it take for 60.0 J of thermal energy to be produced in the 10.0Ω resistor?

Figure P26.63



26.64 •• In the circuit shown in **Fig. P26.64**, $\mathcal{E} = 24.0$ V, $R_1 = 6.00 \Omega$, $R_3 = 12.0 \Omega$, and R_2 can vary between 3.00Ω and 24.0Ω . For what value of R_2 is the power dissipated by heating element R_1 the greatest? Calculate the magnitude of the greatest power.

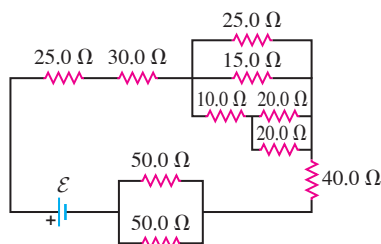
Figure P26.64



26.65 •• (a) Estimate energy usage and cost in your home by creating a table: In the first column list the household appliances that use electricity. In the second column indicate the voltage V used by each appliance. (Use 240 V for the hot-water heater, baseboard heat, and ovens; use 120 V for all other appliances.) In the third column list the current I drawn by each appliance. (This is readily obtained on the Internet.) In the fourth column indicate the number of daily hours T each appliance is used. In the fifth column compute the power of each appliance $P = IV$, and in the sixth column compute the daily energy use $U = PT$, in kilowatt-hours. At the bottom of the final column, sum the total kilowatt-hours used by these appliances. (b) Compute the monthly cost of electricity by multiplying the total daily kilowatt-hours by the cost ($\$0.12/\text{kW} \cdot \text{h}$ is typical). Then multiply the result by 30 days to obtain an estimate of your monthly cost of electricity.

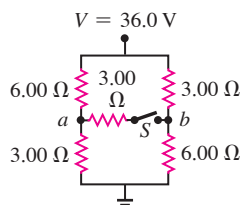
26.66 ••• In the circuit shown in **Fig. P26.66** all the resistors are rated at a maximum power of 2.00 W. What is the maximum emf $\mathcal{E}$ that the battery can have without burning up any of the resistors?

Figure P26.66



26.67 • **Figure P26.67** employs a convention often used in circuit diagrams. The battery (or other power supply) is not shown explicitly. It is understood that the point at the top, labeled “36.0 V,” is connected to the positive terminal of a 36.0 V battery having negligible internal resistance, and that the ground symbol at the bottom is connected to the negative terminal of the battery. The circuit is completed through the battery, even though it is not shown. (a) What is the potential difference V_{ab} , the potential of point a relative to point b , when the switch S is open? (b) What is the current through S when it is closed? (c) What is the equivalent resistance when S is closed?

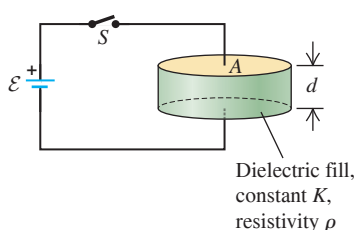
Figure P26.67



26.68 •• Three identical resistors are connected in series. When a certain potential difference is applied across the combination, the total power dissipated is 45.0 W. What power would be dissipated if the three resistors were connected in parallel across the same potential difference?

26.69 ••• **CP** Two capacitor plates with area A are separated by a distance d . The space between the plates is filled with dielectric material with dielectric constant K and resistivity ρ . This capacitor is attached to a battery that supplies a constant potential $\mathcal{E}$, as shown in **Fig. P26.69**, and is fully charged. At time $t = 0$ the switch S is opened. Owing to the nonzero value of ρ , this capacitor discharges by leaking. We can model this device as a capacitor and a resistor in parallel. (a) Analyze this circuit and determine the time constant, characterizing the discharge in terms of the parameters given above. Now assume the capacitor has plates with area $A = 1.00 \text{ cm}^2$ separated by $d = 30.0 \mu\text{m}$ and is filled with a ceramic of dielectric constant $K = 8.70$ and resistivity $\rho = 3.10 \times 10^{12} \Omega \cdot \text{m}$.

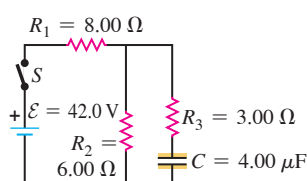
Figure P26.69



(b) If $\mathcal{E} = 5.00 \text{ V}$, what is the charge on the capacitor at $t = 0$? (c) At what time will the capacitor have half of its original charge? (d) What is the magnitude of the leaking current at that point?

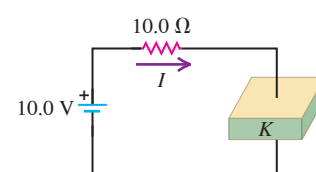
26.70 • The capacitor in **Fig. P26.70** is initially uncharged. The switch S is closed at $t = 0$. (a) Immediately after the switch is closed, what is the current through each resistor? (b) What is the final charge on the capacitor?

Figure P26.70



26.71 ••• **CP CALC** An R - C circuit is constructed as shown in **Fig. P26.71**. The capacitor has $2.00 \text{ cm} \times 2.00 \text{ cm}$ square plates that are parallel and separated by a distance of 3.00 mm, and it is fully charged. At time $t = 0$, a $2.00 \text{ cm} \times 2.00 \text{ cm} \times 3.00 \text{ mm}$ slab with dielectric constant $K = 12.0$ is slid between the plates as shown. (a) How much charge is on the upper plate before the slab is inserted? (b) What is the current I immediately after the slab is inserted? (c) How much energy is stored in the capacitor immediately after the slab is inserted? (d) By how much does the energy in the capacitor increase from that value after a long time has passed? (e) How much total energy is eventually supplied by the battery after the slab is inserted? (*Hint*: Determine the current as a function of time; then integrate the power delivered by the battery, $\int_0^\infty VI(t) dt$.) (f) How much energy is dissipated by the resistor?

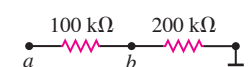
Figure P26.71



26.72 ••• A $6.00 \mu\text{F}$ capacitor that is initially uncharged is connected in series with a 5.00Ω resistor and an emf source with $\mathcal{E} = 50.0 \text{ V}$ and negligible internal resistance. At the instant when the resistor is dissipating electrical energy at a rate of 300 W, how much energy has been stored in the capacitor?

26.73 • Point a in **Fig. P26.73** is maintained at a constant potential of 400 V above ground. (See Problem 26.67.)

Figure P26.73

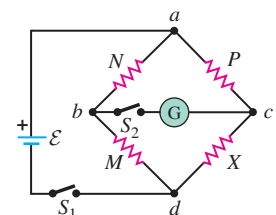


(a) What is the reading of a voltmeter with the proper range and with resistance $5.00 \times 10^4 \Omega$ when connected between point b and ground? (b) What is the reading of a voltmeter with resistance $5.00 \times 10^6 \Omega$? (c) What is the reading of a voltmeter with infinite resistance?

26.74 •• **The Wheatstone Bridge.**

Figure P26.74

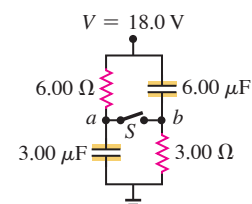
The circuit shown in **Fig. P26.74**, called a *Wheatstone bridge*, is used to determine the value of an unknown resistor X by comparison with three resistors M , N , and P whose resistances can be varied. For each setting, the resistance of each resistor is precisely known. With switches S_1 and S_2 closed, these resistors are varied until the current in the galvanometer G is zero; the bridge is then said to be *balanced*.



(a) Show that under this condition the unknown resistance is given by $X = MP/N$. (This method permits very high precision in comparing resistors.) (b) If galvanometer G shows zero deflection when $M = 850.0 \Omega$, $N = 15.00 \Omega$, and $P = 33.48 \Omega$, what is the unknown resistance X ?

26.75 • (See Problem 26.67.) (a) What is the potential of point a with respect to point b in **Fig. P26.75** when the switch S is open? (b) Which point, a or b , is at the higher potential? (c) What is the final potential of point b with respect to ground when S is closed? (d) How much does the charge on each capacitor change when S is closed?

Figure P26.75

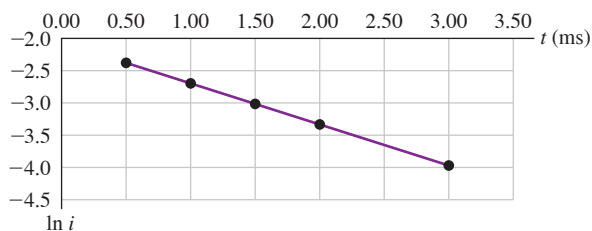


26.76 • A resistor with $R = 850 \Omega$ is connected to the plates of a charged capacitor with capacitance $C = 4.62 \mu\text{F}$. Just before the connection is made, the charge on the capacitor is 6.90 mC. (a) What is the energy initially stored in the capacitor? (b) What is the electrical power dissipated in the resistor just after the connection is made? (c) What is the electrical power dissipated in the resistor at the instant when the energy stored in the capacitor has decreased to half the value calculated in part (a)?

26.77 • A capacitor that is initially uncharged is connected in series with a resistor and an emf source with $\mathcal{E} = 110 \text{ V}$ and negligible internal resistance. Just after the circuit is completed, the current through the resistor is $6.5 \times 10^{-5} \text{ A}$. The time constant for the circuit is 5.2 s . What are the resistance of the resistor and the capacitance of the capacitor?

26.78 •• DATA You set up the circuit shown in Fig. 26.22a, where $R = 196 \Omega$. You close the switch at time $t = 0$ and measure the magnitude i of the current in the resistor R as a function of time t since the switch was closed. Your results are shown in Fig. P26.78, where you have chosen to plot $\ln i$ as a function of t . (a) Explain why your data points lie close to a straight line. (b) Use the graph in Fig. P26.78 to calculate the capacitance C and the initial charge Q_0 on the capacitor. (c) When $i = 0.0500 \text{ A}$, what is the charge on the capacitor? (d) When $q = 0.500 \times 10^{-4} \text{ C}$, what is the current in the resistor?

Figure P26.78



26.79 •• DATA You set up the circuit shown in Fig. 26.20, where $C = 5.00 \times 10^{-6} \text{ F}$. At time $t = 0$, you close the switch and then measure the charge q on the capacitor as a function of the current i in the resistor. Your results are given in the table:

$i \text{ (mA)}$	56.0	48.0	40.0	32.0	24.0
$q \text{ (}\mu\text{C)}$	10.1	19.8	30.2	40.0	49.9

(a) Graph q as a function of i . Explain why the data points, when plotted this way, fall close to a straight line. Find the slope and y-intercept of the straight line that gives the best fit to the data. (b) Use your results from part (a) to calculate the resistance R of the resistor and the emf $\mathcal{E}$ of the battery. (c) At what time t after the switch is closed is the voltage across the capacitor equal to 10.0 V ? (d) When the voltage across the capacitor is 4.00 V , what is the voltage across the resistor?

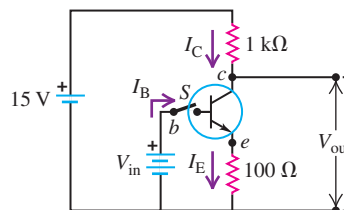
26.80 •• DATA The electronics supply company where you work has two different resistors, R_1 and R_2 , in its inventory, and you must measure the values of their resistances. Unfortunately, stock is low, and all you have are R_1 and R_2 in parallel and in series—and you can't separate these two resistor combinations. You separately connect each resistor network to a battery with emf 48.0 V and negligible internal resistance and measure the power P supplied by the battery in both cases. For the series combination, $P = 48.0 \text{ W}$; for the parallel combination, $P = 256 \text{ W}$. You are told that $R_1 > R_2$. (a) Calculate R_1 and R_2 . (b) For the series combination, which resistor consumes more power, or do they consume the same power? Explain. (c) For the parallel combination, which resistor consumes more power, or do they consume the same power?

CHALLENGE PROBLEMS

26.81 ••• The three-terminal device circled in Fig. P26.81 is an NPN transistor. It works using three simple rules: I. The net current into the device is the same as the net current out of the device, so $I_C + I_B = I_E$. II. The potential at point e is always 0.60 V less than the potential at point b . Thus, $V_{in} - V_e = 0.60 \text{ V}$. III. The current into point c is always a fixed multiple of the current into point b , so $I_C = \beta I_B$, where $\beta \gg 1$ is a parameter characteristic of the transistor. Use these rules to answer the questions. (a) What is the potential V_{out} in terms of V_{in} , in the limit $\beta \rightarrow \infty$? (Hint: Determine I_E in terms of V_{in} ,

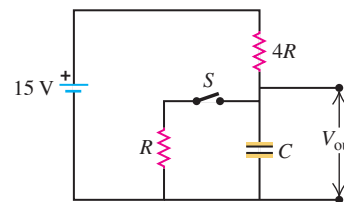
and use this to determine I_C in terms of V_{in} and β . The potential difference across the $1.0 \text{ k}\Omega$ resistor can then be used to determine V_{out} . Take the limit $\beta \rightarrow \infty$ in this result.) (b) What value of V_{in} is needed so that $V_{out} = 7.5 \text{ V}$? (c) If the input potential includes a small time-dependent "signal," so that $V_{in} = 15.0 \text{ V} + v_{in}(t)$, then the output potential is $V_{out} = 7.5 \text{ V} + Gv_{in}(t)$, where G is a "gain" factor. What is G for this circuit?

Figure P26.81



26.82 ••• In the circuit shown in Fig. P26.82, the switch S has been closed for a long time. (a) What is the output voltage V_{out} ? Now the switch is opened and the output voltage increases. When the output voltage reaches 10.0 V , the switch closes and then the capacitor begins to discharge. When the output voltage reaches 5.0 V , the switch opens again. Thus this circuit undergoes periodic cycles of charging and discharging. (b) What is the time constant during a charging phase? (c) What is the time constant during a discharging phase? (d) What length of time passes between subsequent 10.0 V output voltages? (e) If $R = 10.0 \text{ k}\Omega$ and $C = 10.0 \mu\text{F}$, what is the frequency of operation? (Note: This circuit is a simplified example of a timer circuit, a crucial element in many electronics applications.)

Figure P26.82

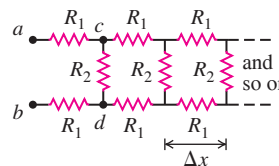


26.83 ••• An Infinite Network. As shown in Fig. P26.83, a network of resistors of resistances R_1 and R_2 extends to infinity toward the right. Prove that the total resistance R_T of the infinite network is equal to

$$R_T = R_1 + \sqrt{R_1^2 + 2R_1R_2}$$

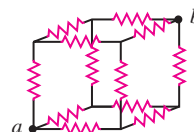
(Hint: Since the network is infinite, the resistance of the network to the right of points c and d is also equal to R_T .)

Figure P26.83



26.84 ••• Suppose a resistor R lies along each edge of a cube (12 resistors in all) with connections at the corners. Find the equivalent resistance between two diagonally opposite corners of the cube (points a and b in Fig. P26.84).

Figure P26.84



26.85 ••• BIO Attenuator Chains and Axons. The infinite network of resistors shown in Fig. P26.83 is known as an *attenuator chain*, since this chain of resistors causes the potential difference between the upper and lower wires to decrease, or attenuate, along the length of the chain. (a) Show that if the potential difference between the points a and b in Fig. 26.83 is V_{ab} , then the potential difference between points c and d is $V_{cd} = V_{ab}/(1 + \beta)$, where $\beta = 2R_1(R_T + R_2)/R_T R_2$ and R_T , the total resistance of the network, is given in Challenge Problem 26.83. (See the hint given in that problem.) (b) If the potential difference between terminals a and b at the left end of the infinite network is V_0 , show that the potential difference between the upper and lower wires n segments from the left end is $V_n = V_0/(1 + \beta)^n$. If $R_1 = R_2$, how many segments are needed to decrease the potential difference V_n to less than 1.0% of V_0 ? (c) An infinite attenuator chain provides a model of the propagation of a voltage pulse along a nerve fiber, or axon. Each segment of the network in Fig. P26.83 represents a short segment of the axon of length Δx . The resistors R_1 represent the resistance of the fluid inside and outside the membrane wall of the axon. The resistance of the membrane to current flowing through the wall is represented by R_2 . For an axon segment of length $\Delta x = 1.0 \mu\text{m}$, $R_1 = 6.4 \times 10^3 \Omega$ and $R_2 = 8.0 \times 10^8 \Omega$ (the membrane wall is a good insulator). Calculate the total resistance R_T and β for an infinitely long axon. (This is a good approximation, since the length of an axon is much greater than its width; the largest axons in the human nervous system are longer than 1 m but only about 10^{-7} m in radius.) (d) By what fraction does the potential difference between the inside and outside of the axon decrease over a distance of 2.0 mm? (e) The attenuation of the potential difference calculated in part (d) shows that the axon cannot simply be a passive, current-carrying electrical cable; the potential difference must periodically be reinforced along the axon's length. This reinforcement mechanism is slow, so a signal propagates along the axon at only about 30 m/s. In situations where faster response is required, axons are covered with a segmented sheath of fatty myelin. The segments are about 2 mm long, separated by gaps called the *nodes of Ranvier*. The myelin increases the resistance of a

$1.0\text{-}\mu\text{m}$ -long segment of the membrane to $R_2 = 3.3 \times 10^{12} \Omega$. For such a myelinated axon, by what fraction does the potential difference between the inside and outside of the axon decrease over the distance from one node of Ranvier to the next? This smaller attenuation means the propagation speed is increased.

MCAT-STYLE PASSAGE PROBLEMS

BIO Nerve Cells and R-C Circuits. The portion of a nerve cell that conducts signals is called an *axon*. Many of the electrical properties of axons are governed by ion channels, which are protein molecules that span the axon's cell membrane. When open, each ion channel has a pore that is filled with fluid of low resistivity and connects the interior of the cell electrically to the medium outside the cell. In contrast, the lipid-rich cell membrane in which ion channels reside has very high resistivity.

26.86 Assume that a typical open ion channel spanning an axon's membrane has a resistance of $1 \times 10^{11} \Omega$. We can model this ion channel, with its pore, as a 12-nm-long cylinder of radius 0.3 nm. What is the resistivity of the fluid in the pore? (a) $10 \Omega \cdot \text{m}$; (b) $6 \Omega \cdot \text{m}$; (c) $2 \Omega \cdot \text{m}$; (d) $1 \Omega \cdot \text{m}$.

26.87 In a simple model of an axon conducting a nerve signal, ions move across the cell membrane through open ion channels, which act as purely resistive elements. If a typical current density (current per unit cross-sectional area) in the cell membrane is 5 mA/cm^2 when the voltage across the membrane (the *action potential*) is 50 mV, what is the number density of open ion channels in the membrane? (a) $1/\text{cm}^2$; (b) $10/\text{cm}^2$; (c) $10/\text{mm}^2$; (d) $100/\mu\text{m}^2$.

26.88 Cell membranes across a wide variety of organisms have a capacitance per unit area of $1 \mu\text{F/cm}^2$. For the electrical signal in a nerve to propagate down the axon, the charge on the membrane "capacitor" must change. What time constant is required when the ion channels are open? (a) $1 \mu\text{s}$; (b) $10 \mu\text{s}$; (c) $100 \mu\text{s}$; (d) 1 ms.

ANSWERS

Chapter Opening Question ?

(ii) The potential difference V is the same across resistors connected in parallel. However, there is a different current I through each resistor if the resistances R are different: $I = V/R$.

Key Example VARIATION Problems

VP26.2.1 (a) 7.00Ω (b) 0.571Ω (c) 2.33Ω (d) 0.857Ω

VP26.2.2 (a) 8.18 A (b) 6.00 A (c) 2.18 A (d) 2.18 A

VP26.2.3 (a) 54.6 W (b) 20.6 W (c) 18.0 W (d) 16.0 W

VP26.2.4 (a) 9.84 W (b) 5.98 W (c) 2.08 W (d) 1.78 W

VP26.7.1 (a) 1 A (b) 7 V (c) 12 W for 12 V battery, 4 W for 4 V battery

VP26.7.2 (a) 7.75 V (b) 1.25 V (c) 0.926 V

VP26.7.3 (a) 33.4 W (b) 18.8 W (c) 2.09 W (d) 52.2 W (e) 25.0 W

VP26.7.4 (a) $(2\mathcal{E}_1 + \mathcal{E}_2)/3R$ (b) $-(\mathcal{E}_1 + 2\mathcal{E}_2)/3R$ (c) $(\mathcal{E}_1 - \mathcal{E}_2)/3R$

VP26.13.1 (a) $2.88 \times 10^{-5} \text{ C}$ (b) $9.00 \times 10^{-7} \text{ A}$ (c) 32.0 s (d) 0.430 (e) 0.570

VP26.13.2 (a) 11.0 s (b) $1.36 \times 10^{-7} \text{ A}$

VP26.13.3 (a) 1.32 M Ω (b) $5.21 \times 10^{-7} \text{ A}$ (c) $1.04 \times 10^{-7} \text{ A}$

VP26.13.4 (a) $6.12 \times 10^{-5} \text{ C}$ (b) $9.89 \times 10^{-6} \text{ C}$

Bridging Problem

(a) 9.39 J (b) $2.02 \times 10^4 \text{ W}$ (c) $4.65 \times 10^{-4} \text{ s}$ (d) $7.43 \times 10^3 \text{ W}$